Jellyfin + SWAG + EasyEPG + Telerising on UGREEN NAS (Docker)

Manual (EN) – Version v1.0 – As of 01 Feb. 2026

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1. Introduction

This package provides a complete Jellyfin server environment as a Docker Compose project. It includes:

- Jellyfin Media Server (streaming, user management, libraries).
- SWAG (reverse proxy with Let's Encrypt certificates and optional Fail2Ban).
- EasyEPG (EPG generation / XMLTV output).
- Telerising (M3U/streaming API as a source for Live TV).

The goal is easy deployment via the UGREEN Docker app (Projects), including centralized configuration via a .env file.

2. Features

- Jellyfin reachable on the LAN (HTTP); optional internet access via SWAG (HTTPS).
- Certificates automatically via Let's Encrypt (SWAG).
- Optional listener for Jellyfin SSL (port 8920) via SWAG.
- Hardware transcoding via Intel iGPU (Quick Sync/QSV) or on ARM/Rockchip via RKMPP (see section 7.2).
- EasyEPG creates XMLTV files that can later be used in Jellyfin as an EPG source.
- Telerising provides Live TV streams as M3U/HTTP endpoints (setup later in this manual).
- Central variables (paths, ports, domain, UID/GID) in a .env file.

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3. Package contents and folder structure

Included in the package (excerpt):

- docker-composeDXP.yaml - Starts Jellyfin, SWAG, EasyEPG and Telerising (DXP series / Intel QSV).
- docker-composeDH.yaml - Starts Jellyfin, SWAG, EasyEPG and Telerising (DH series / Rockchip RKMPP).
- .env - Central variables (paths, ports, domain, UID/GID).
- linuxserver_swag/config - SWAG configuration incl. Nginx proxy-confs.
- takealug_new-easyepg - EasyEPG data and XML output folder.
- telerising - Telerising data directory.

Note: Before importing in the Docker app, you must rename the matching compose file (DXP or DH) to docker-compose.yaml. You can keep the other file in the folder as a template.

Recommended location is a folder on a data volume, e.g. under:

```
/volume2/docker/JellyfinServer/
```

4. Requirements

- UGREEN NAS with UGOS Pro and Docker/Container Manager (Compose/Projects).
- SMB access to the NAS (to copy the project folder).
- A public domain (e.g. jellyfin.your-domain.tld) and access to DNS settings.
- Router with port forwarding (example: FRITZ!Box).

For hardware transcoding: suitable hardware (Intel iGPU for QSV or Rockchip/ARM for RKMPP) and correct device passthrough into the container (see section 7.2).

5. Preparation: shares and paths

5.1 Ports and port forwarding (example FRITZ!Box)

This manual uses the following example setup for router port forwarding (Internet → NAS):

- TCP 443 → NAS port 50443 (HTTPS / SWAG).
- TCP 80 → NAS port 5080 (HTTP / SWAG).
- TCP 8920 → NAS port 8920 (Jellyfin SSL via SWAG listener, optional).

NAS → container (Docker port mapping from the compose):

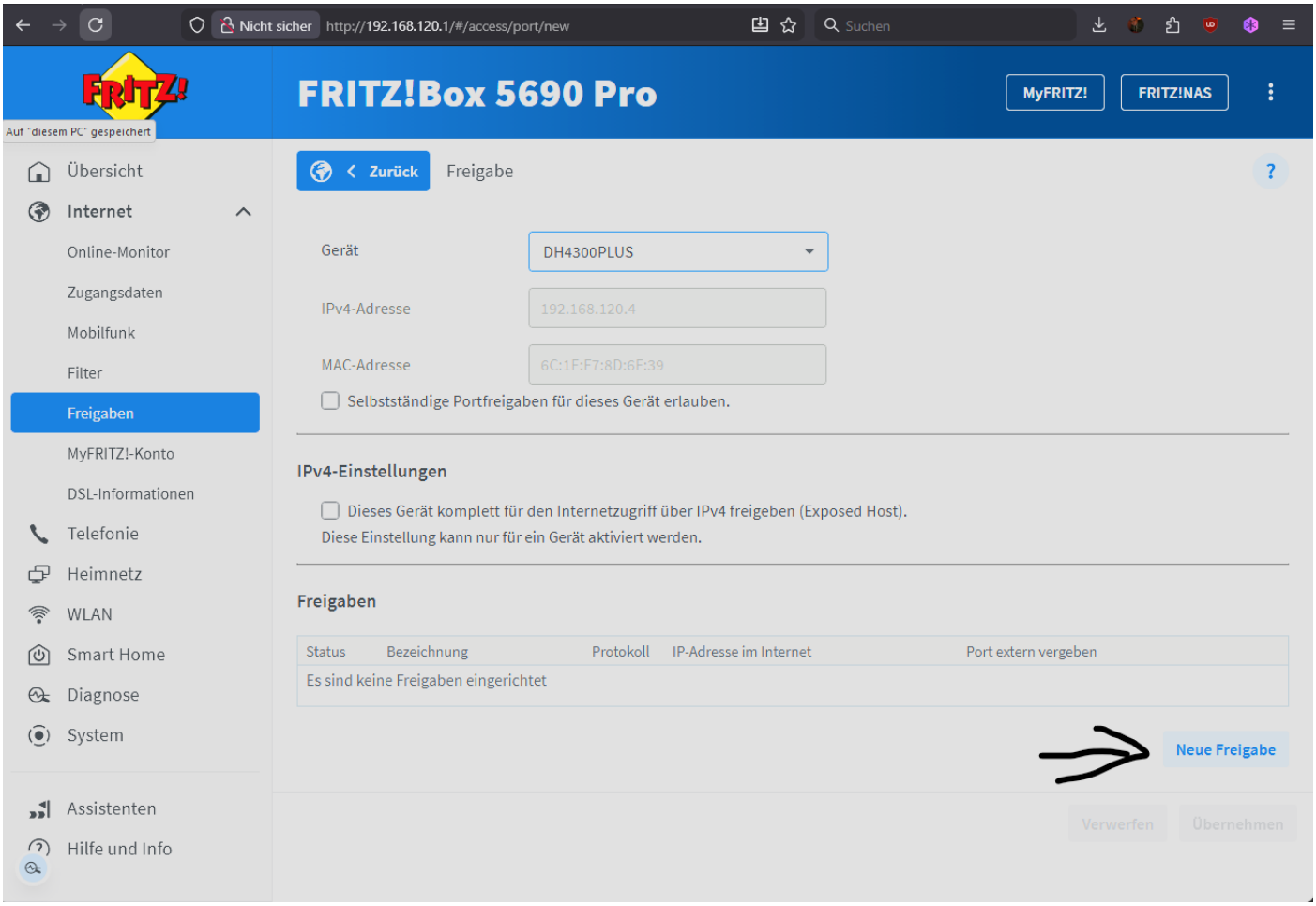
- NAS 50443 → container 443 (SWAG HTTPS).
- NAS 5080 → container 80 (SWAG HTTP).
- NAS 8920 → container 8920 (SWAG listener for Jellyfin SSL).

Important:

- UDP 7359 (discovery) is for LAN only and is not forwarded to the internet.
- The Jellyfin HTTP port 8096 usually stays internal (LAN) and is not published to the internet via port forwarding.

5.1.1 Create FRITZ!Box port forwards (quick guide)

- FRITZ!Box: Internet → Permit Access → Port Sharing → Add Device.
- Select the NAS (or the IP address).
- Port sharing for HTTPS: external 443 → internal 50443 (TCP).
- Port sharing for HTTP: external 80 → internal 5080 (TCP).
- Optional port sharing for Jellyfin SSL: external 8920 → internal 8920 (TCP).



5.1.2 SWAG configuration: subdomain and Fail2Ban (optional)

If you use a different subdomain than “jellyfin”, you must adjust it in two places so that the URL and, if used, Fail2Ban work correctly.

- Adjust the subdomain in the Nginx proxy configuration (server_name):

```
JellyfinServer/linuxserver_swag/config/nginx/proxy-confs/jellyfin.subdomain.conf
```

Open the file ./linuxserver_swag/config/nginx/proxy-confs/jellyfin.subdomain.conf and adjust the “server_name” line to your subdomain (example):

```
server_name jellyfin.*;  # default
server_name YOUR-SUBDOMAIN.*;  # e.g. media.* or tv.*
```

Save the file and then restart the SWAG container (or the project) so the change becomes active.

- Fail2Ban: Open ./linuxserver_swag/config/fail2ban/jail.local and adjust “ignoreip” to your LAN (so you don't lock yourself out):

```
JellyfinServer/linuxserver_swag/config/fail2ban/jail.local
```

Adjust the following networks to your home network if needed (example values):

```
ignoreip = 10.0.0.0/8
           192.168.0.0/16
           172.16.0.0/12
```

In the same file, further down, you will find the recidive block. Here too, you should adjust the “ignoreip” line to your LAN (example):

```
# ==============================
# Recidive (ban repeat offenders longer)
# ==============================
[recidive]
enabled  = true
filter   = recidive
# SWAG writes the fail2ban log here:
logpath  = /config/log/fail2ban/fail2ban.log
bantime  = 7d
findtime = 1d
maxretry = 3
# Optional: to be safe, never ban your LAN
ignoreip = 127.0.0.1/8 192.168.120.0/24
# Recidive should ban "hard" (all ports)
banaction = iptables-allports
```

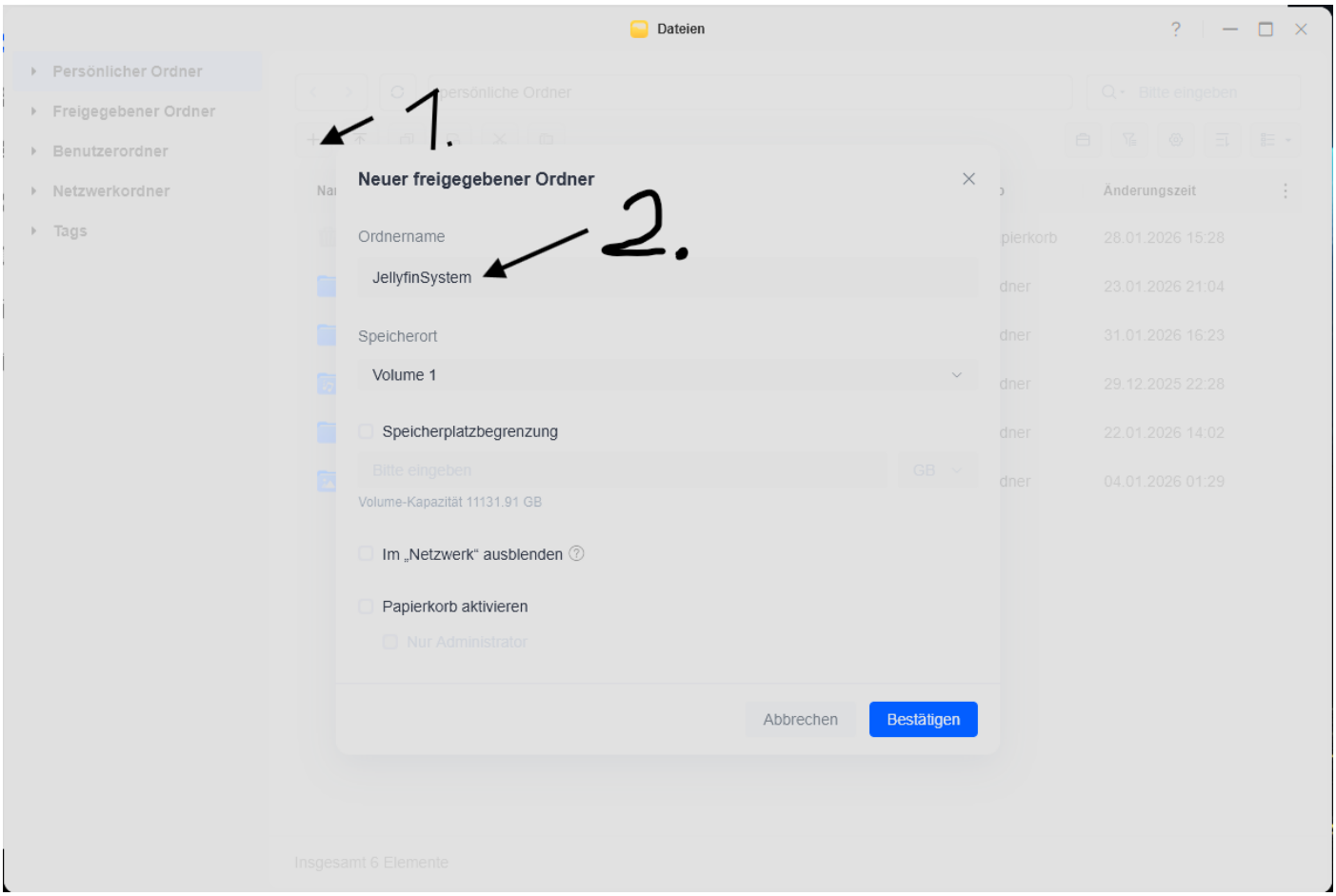
Tip: After changes to jail.local, restart the SWAG container so Fail2Ban reloads the configuration.

5.2 Folders and shares

Create the following folders/shares on your NAS (or check that they already exist). Example paths:

- Project folder: /volume2/docker/JellyfinServer (Compose project). Recommendation: NVMe volume (RAID1).
- MEDIA_DIR: /volume1/Jellyfin (media library). Recommendation: clear folder structure (e.g. /Movies and /Series) so Jellyfin recognizes the content correctly.
- JELLYFIN_SYSTEM_DIR: /volume2/JellyfinSystem (system data/exports/transcodes). Recommendation: NVMe volume (RAID1).

Important: The Docker user (PUID/PGID or UID/GID) needs write permissions in the target folders.



6. Installation

6.1 Copy files to the Docker directory via SMB

Copy the complete project folder (including docker-composeDXP.yaml, docker-composeDH.yaml and .env) to the NAS via SMB, e.g. to:

```
/volume2/docker/JellyfinServer/
```

Tip: The .env is located in the same folder as the compose files.

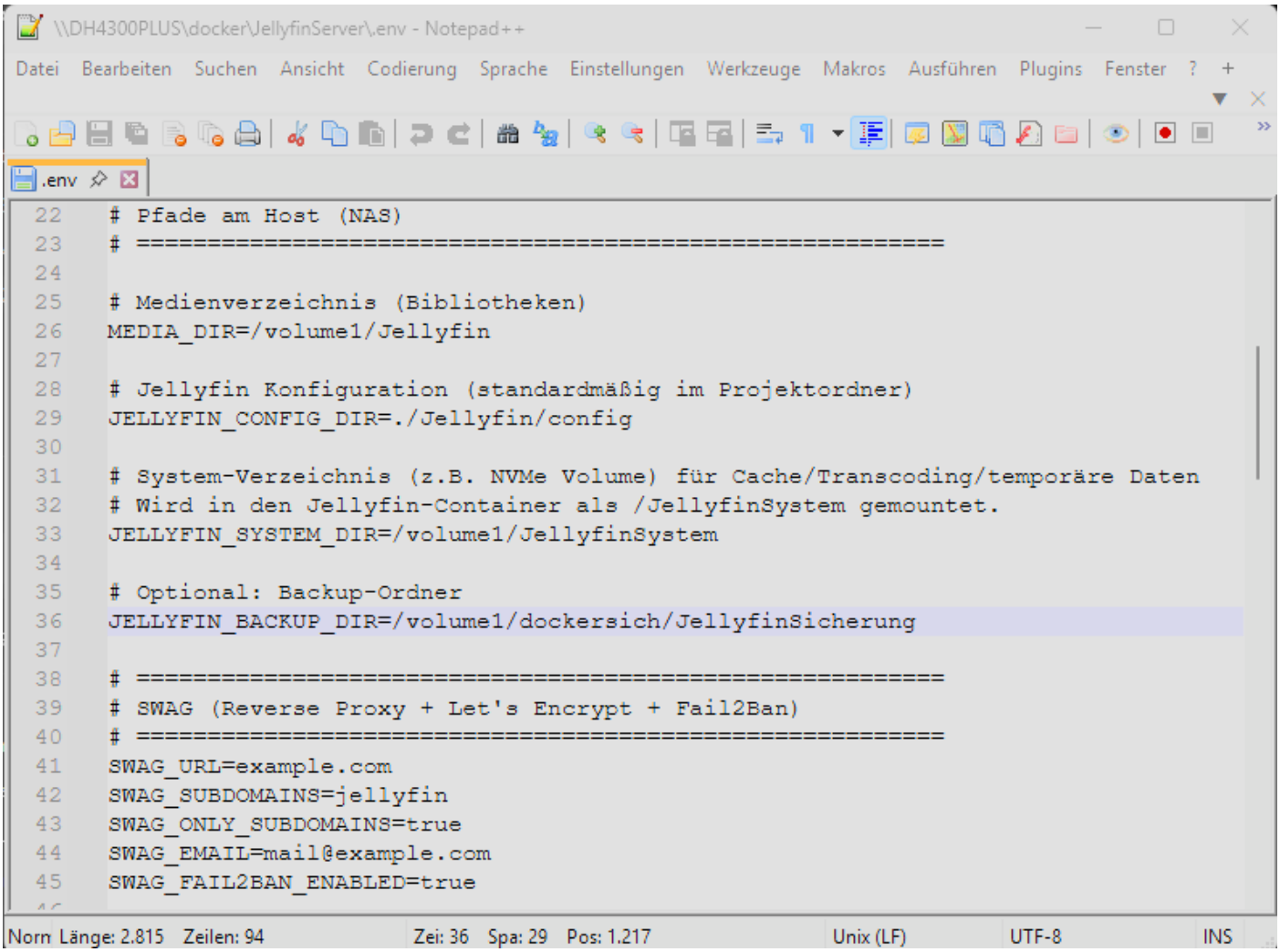
Important: The package contains two compose files. Choose the correct file for your NAS and rename it to docker-compose.yaml before importing.

- DXP series (Intel): docker-composeDXP.yaml -> docker-compose.yaml
- DH series (Rockchip/ARM): docker-composeDH.yaml -> docker-compose.yaml

6.2 Edit .env (required fields)

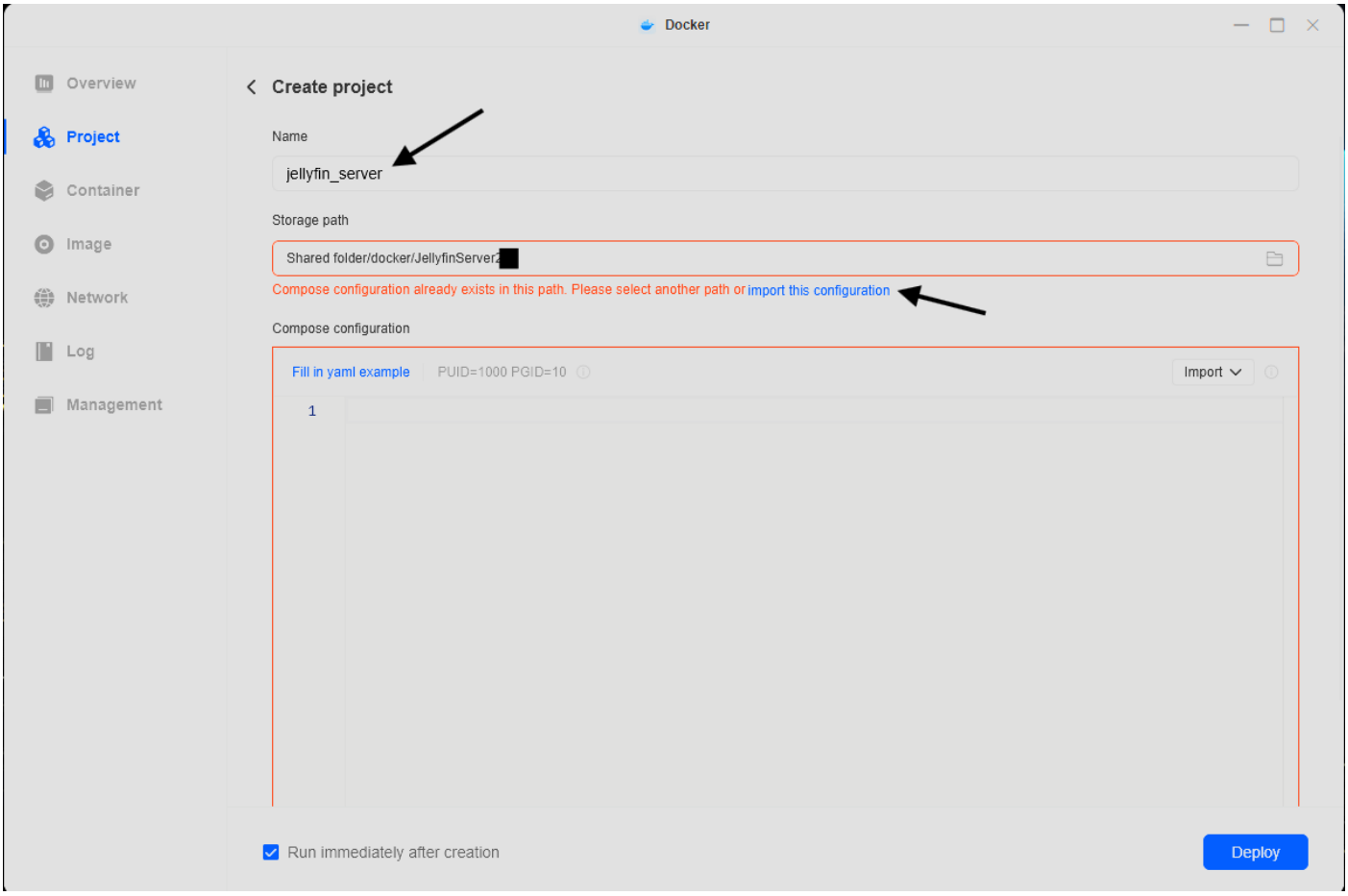
In the .env, adjust at least the following variables:

- MEDIA_DIR - Path to your Jellyfin media share.
- JELLYFIN_SYSTEM_DIR - Base folder for Jellyfin system data (e.g. cache/transcodes/temp).
- SWAG_URL - Your domain (base domain).
- SWAG_EMAIL - Email address for Let's Encrypt.
- Optional (depending on your system): PUID/PGID, ports, subdomains.



6.3 Add the project in the Docker app (UGOS: Docker/Container Manager → Projects)

- Open Docker app → Projects.
- Create a new project and choose Import.
- Select the project folder (contains docker-compose.yaml + .env) and import docker-compose.yaml.
- Make sure the .env is in the same folder.
- Start the project and check status (all containers should be running).



Note: The first start of SWAG can take a little while because certificates are requested and Nginx is configured.

Important (UID/GID): Use the same user and group ID for all containers.

- User ID (same): PUID (SWAG/Jellyfin), USER_ID (EasyEPG), TELERISING_UID (Telerising)
- Group ID (same): PGID (SWAG/Jellyfin), GROUP_ID (EasyEPG), TELERISING_GID (Telerising)

6.4 Check logs

- SWAG logs: important for certificate creation and Nginx errors.
- Jellyfin logs: important for permissions, hardware transcoding, plugins.
- EasyEPG/Telerising logs: later relevant for Live TV setup.

7. Access and initial setup of Jellyfin

Local access (LAN):

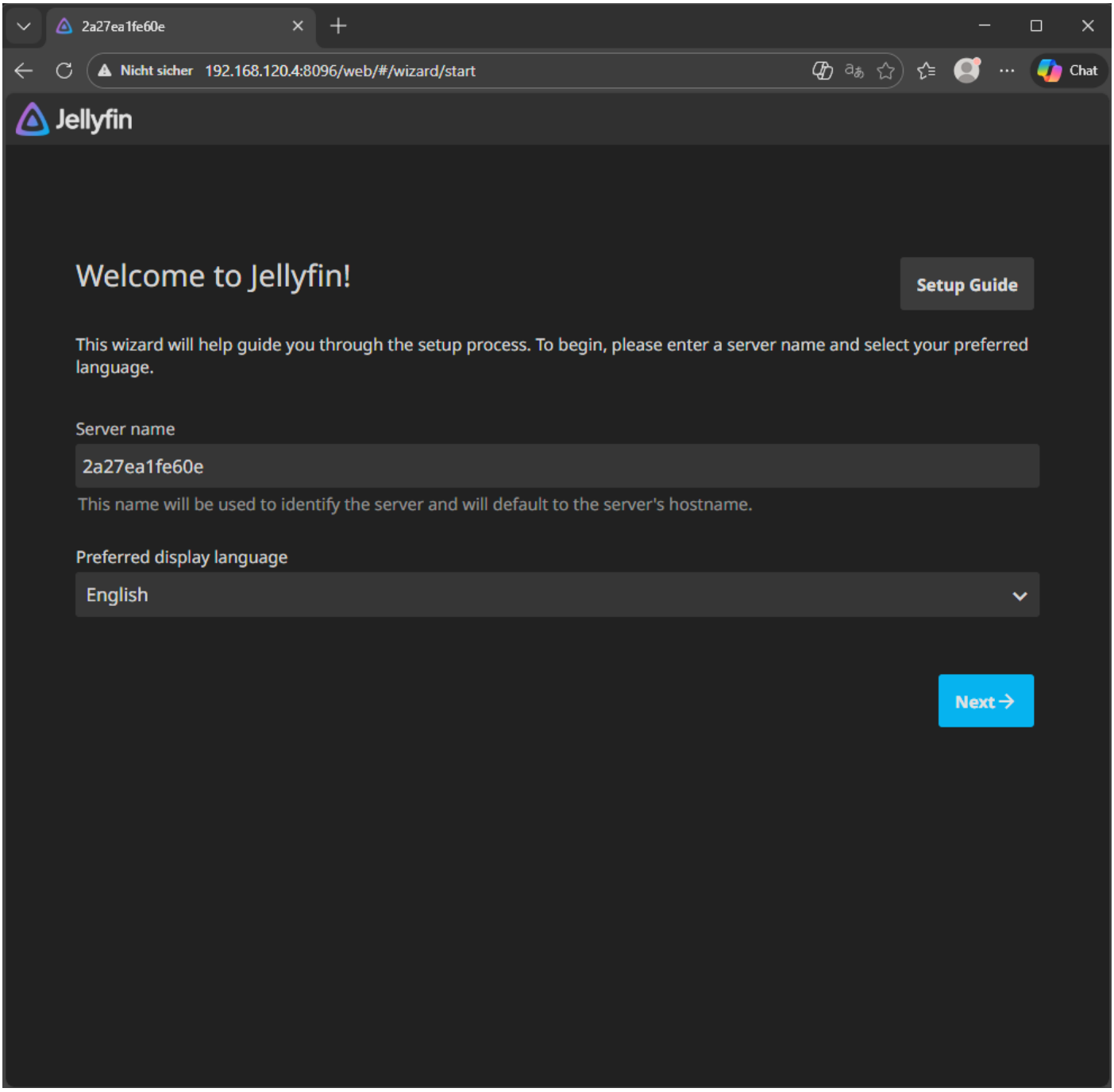
- http://NAS-IP:8096http://NAS-IP:8096 (Jellyfin web UI via HTTP).

External access (internet) via SWAG:

- https://jellyfin.your-domain.tld (default via port 443).

7.1 Initial setup in Jellyfin

- Run through the wizard: language, user, libraries, metadata settings.
- Point libraries to the mounted paths (e.g. /data/media).
- Adjust the server name.



7.2 Set up hardware transcoding

Jellyfin supports hardware transcoding without an additional license. Prerequisite: the hardware is available and the required devices are passed through into the container.

Note (package): For hardware acceleration, this package already prepares everything in the matching compose file depending on your NAS (DXP: docker-composeDXP.yaml with /dev/dri; DH: docker-composeDH.yaml with RKMPP devices and privileged).

DXP 4800 Plus (Intel iGPU): Dashboard → Playback → Transcoding → set hardware acceleration to “Intel Quick Sync (QSV)” (fallback: VA-API). /dev/dri must be mounted in the container.

DH4300 Plus / DH series (Rockchip/ARM): Dashboard → Playback → Transcoding → set hardware acceleration to “Rockchip MPP (RKMPP)” (Jellyfin ≥ 10.9). Note: RKMPP requires additional device configuration depending on the system (MPP/RGA/DMA etc.).

Transcoding temporary path (in Jellyfin): /JellyfinSystem/transcodes (Host: ${JELLYFIN_SYSTEM_DIR}/transcodes).

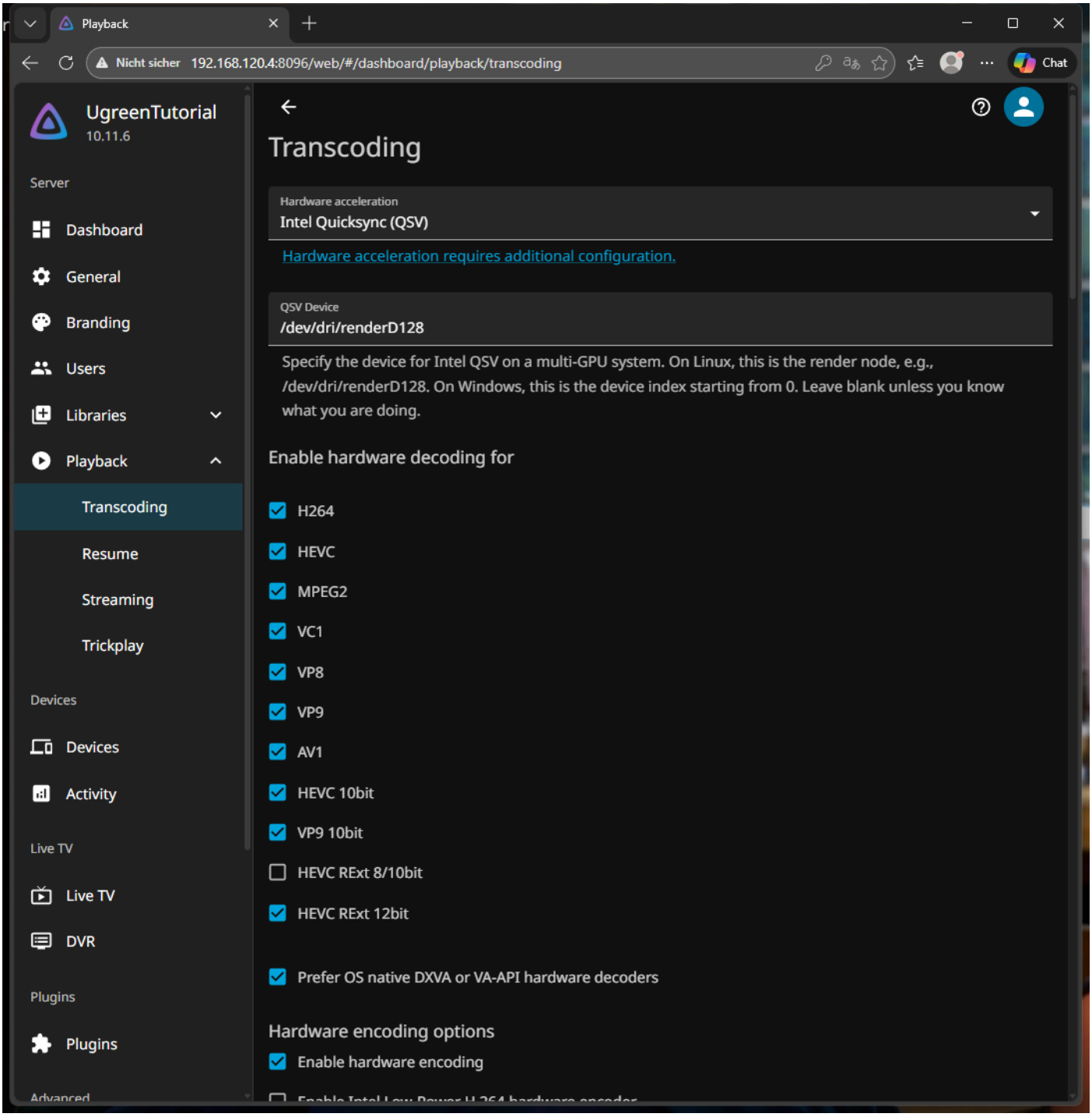
Tip: PGS/ASS subtitles often cause burn-in (software) and greatly increase load; if possible, use SRT and choose remote bitrate so Direct Play is preferred.

7.2.1 Recommended codecs and hardware encoding (DXP 4800 Plus vs. DH4300 Plus)

Open in Jellyfin: Dashboard → Playback → Transcoding.

DXP 4800 Plus (Intel iGPU):

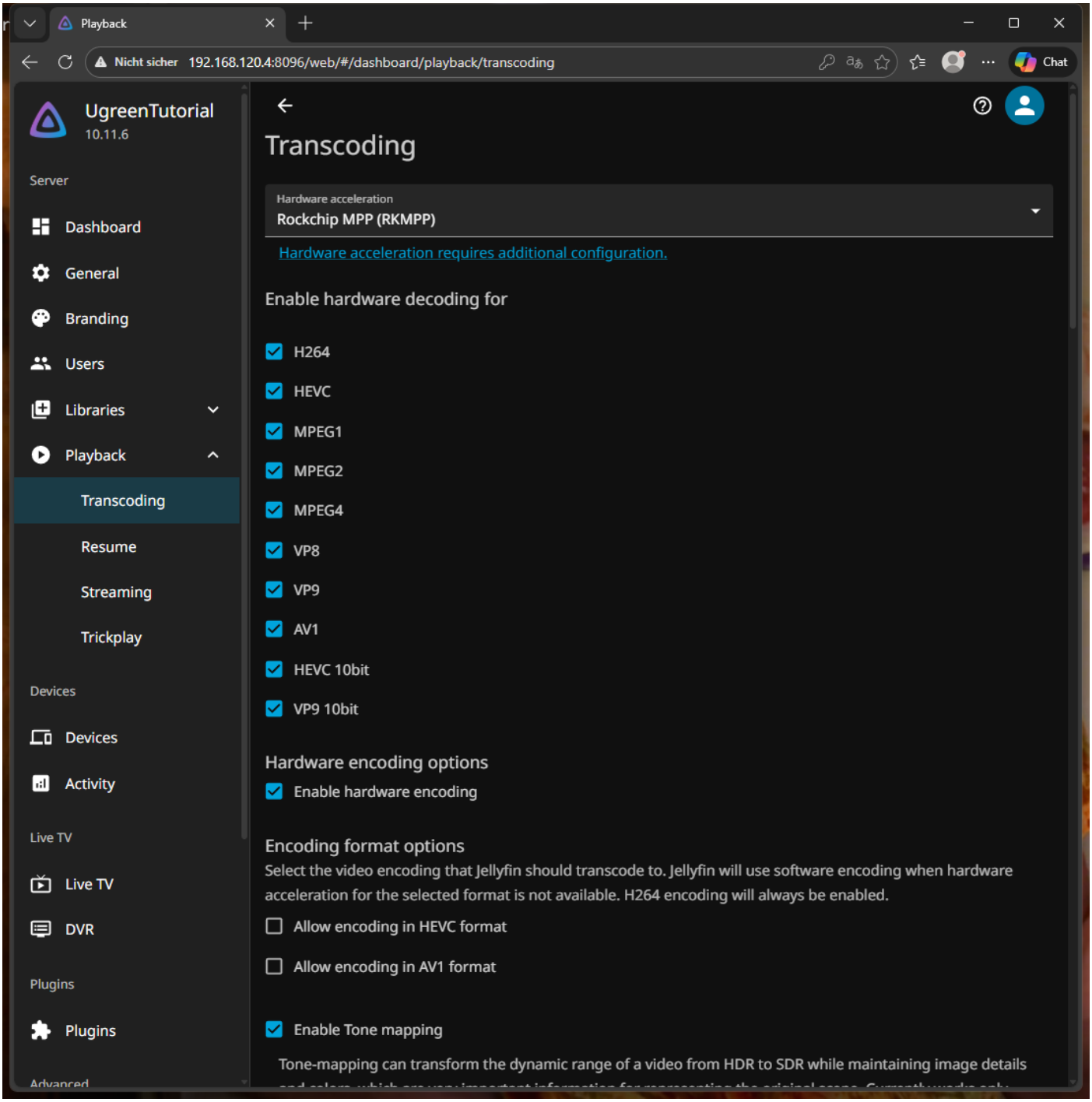
- Hardware acceleration: Intel Quick Sync (QSV) (fallback: VA-API).
- Enable hardware decoding: check all codecs that are shown to you (at least H.264 + HEVC; optional MPEG-2/VC-1/VP9/HEVC 10bit/AV1 depending on what's shown).
- Enable hardware encoding: H.264 (HEVC optional, depending on client compatibility).
- Encoding format options: “Allow encoding in HEVC format” = optional ON; “Allow encoding in AV1 format” = OFF.
- Tone mapping (HDR → SDR): default OFF. Enable only if needed; algorithm: BT.2390 (recommended).


Playback
Nicht sicher 192.168.120.4:8096/web/#/dashboard/playback/transcoding
Chat
UgreenTutorial
10.11.6
Server
Dashboard
General
Branding
Users
Libraries
Playback
Transcoding
Resume
Streaming
Trickplay
Devices
Devices
Activity
Live TV
Live TV
DVR
Plugins
Plugins
Transcoding
Hardware acceleration
Intel Quicksync (QSV)
Hardware acceleration requires additional configuration.
QSV Device
/dev/dri/renderD128
Specify the device for Intel QSV on a multi-GPU system. On Linux, this is the render node, e.g., /dev/dri/renderD128. On Windows, this is the device index starting from 0. Leave blank unless you know what you are doing.
Enable hardware decoding for
H264
HEVC
MPEG2
VC1
VP8
VP9
AV1
HEVC 10bit
VP9 10bit
HEVC RExt 8/10bit
HEVC RExt 12bit
Prefer OS native DXVA or VA-API hardware decoders
Hardware encoding options
Enable hardware encoding

DH4300 Plus / DH series (Rockchip/ARM):

- Hardware acceleration: Rockchip MPP (RKMPP) (Jellyfin ≥ 10.9 required).
- Enable hardware decoding: check all codecs that are shown to you (e.g. H.264, HEVC, MPEG1/2/4, VP8/VP9, HEVC 10bit, VP9 10bit, AV1 – depending on device/build).
- Enable hardware encoding: ON (recommended), but only after testing. If you see artifacts/crashes, switch it OFF again and use decoding only.
- Encoding format options: “Allow encoding in HEVC format” = ON; “Allow encoding in AV1 format” = OFF.
- Tone mapping (HDR → SDR): Enable only if you transcode HDR content for SDR devices. Algorithm: BT.2390 (recommended).

Note: You set the temporary transcoding path in Jellyfin to /JellyfinSystem/transcodes (see below).



7.2.2 Trickplay (optional - preview images when seeking)

Trickplay generates preview images for fast seeking. Depending on the library, this can cost a lot of CPU/IO.

Settings: Dashboard → Playback → Trickplay.

- Scan behavior: Non Blocking (default).
- Process priority: Below Normal (so the NAS doesn't feel sluggish).

DXP 4800 Plus (Intel/QSV):

- Enable hardware decoder: ON.
- Enable hardware-accelerated MJPEG encoding: ON.
- Optional: Generate images from keyframes only: ON (reduces load, timing less accurate).

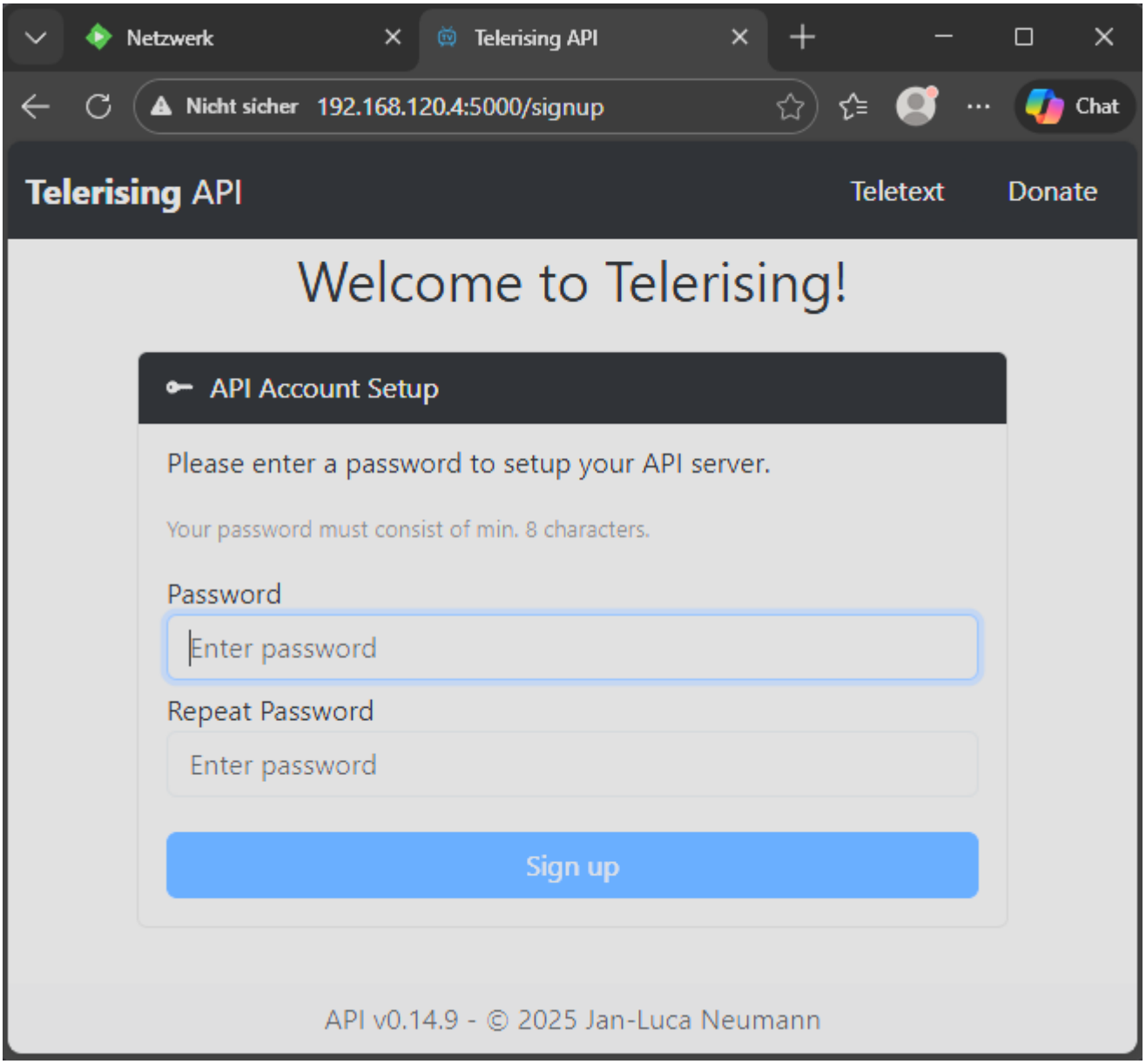
DH series / DH4300 Plus (RKMPP):

- Start without hardware: hardware decoder OFF and MJPEG encoding OFF (most stable option).
- Optional: Generate images from keyframes only: ON (reduces load).
- If RKMPP is proven stable on your system, you can test enabling the hardware decoder. If there are issues, switch it OFF again.

If you don't need Trickplay: just leave it disabled.

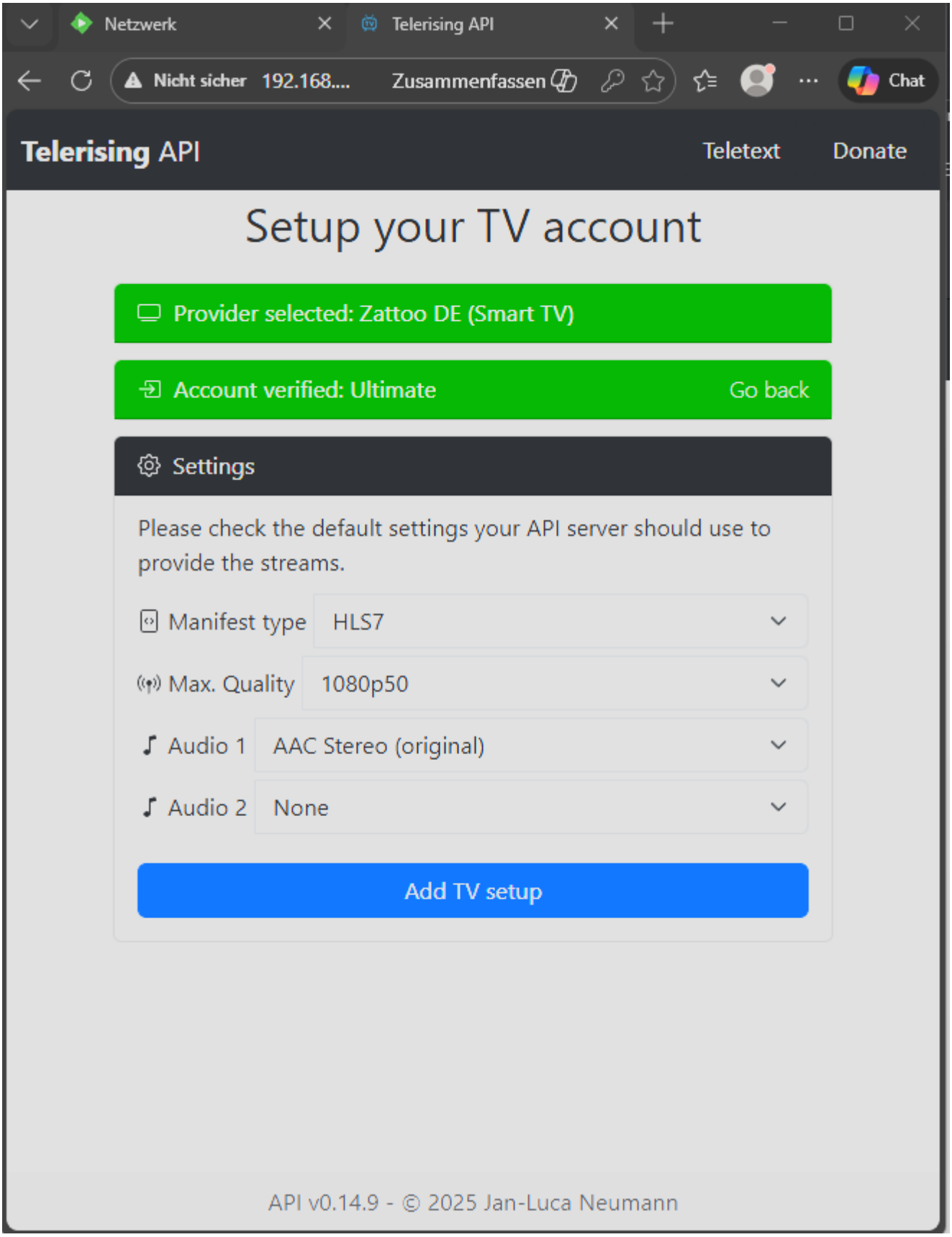
7.3 Set up Telerising

Open Telerising in your browser at http://NAS-IP:5000 (replace NAS-IP with the IP address of your NAS). On first launch, you will be prompted to set a password.http://NAS-IP:5000



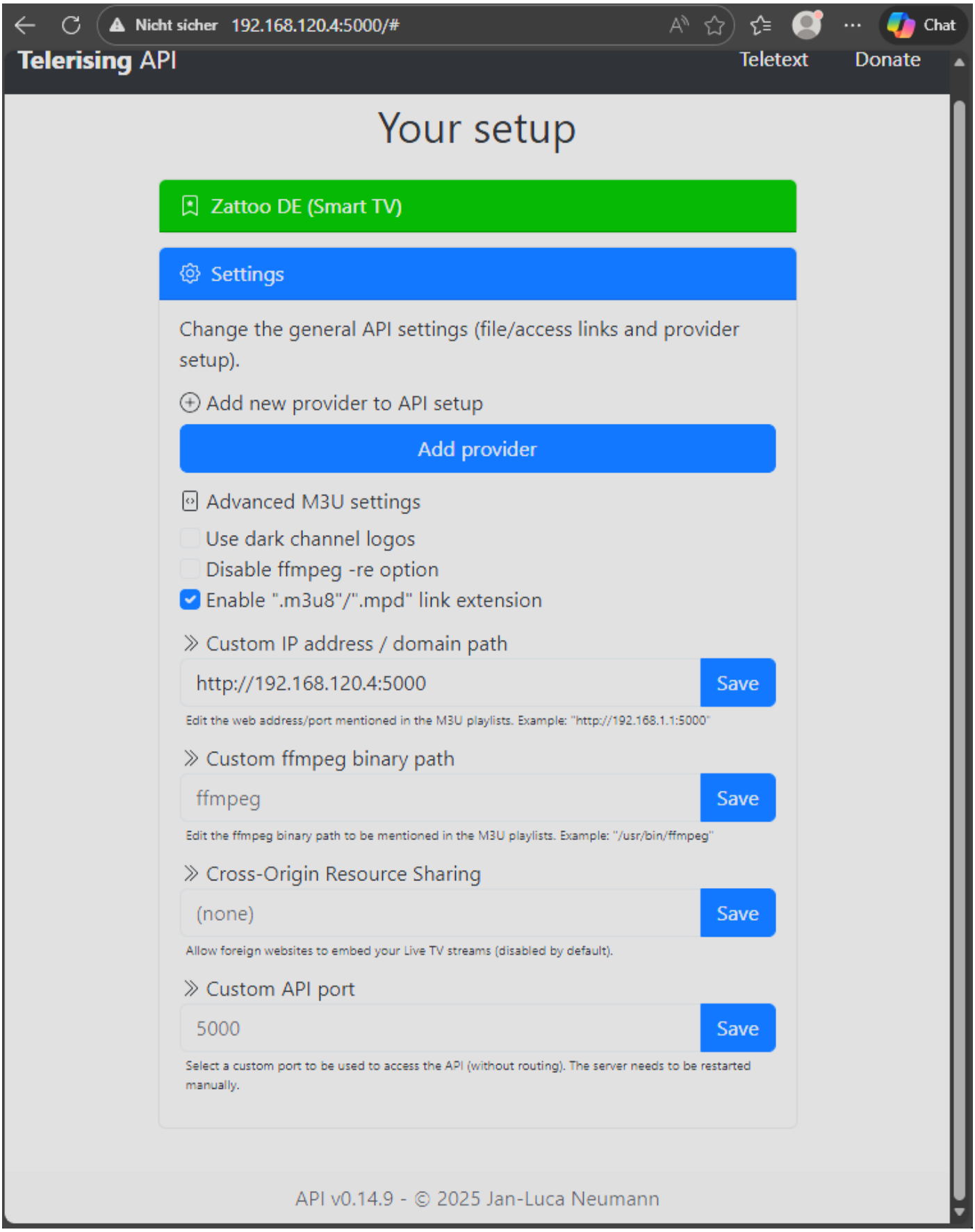
In the next step, set up your Live TV provider. In this example I use Zattoo DE.

After selecting the Live TV provider and entering your credentials, set the manifest type, quality and audio tracks.

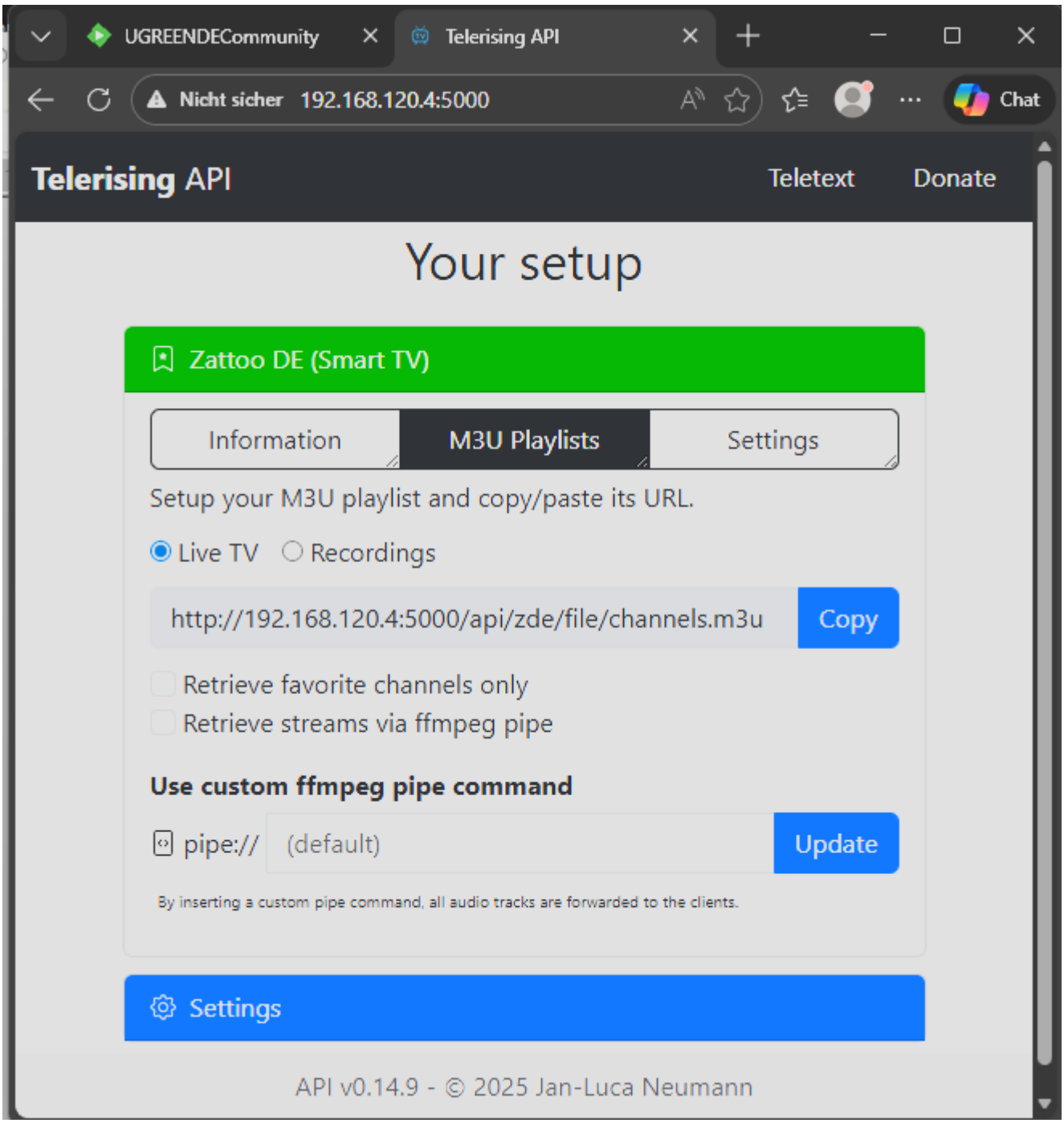


When the selection is complete, click “Add TV setup”.

Now set the IP address so that the later links in Jellyfin are correct.
Example (NAS IP: 192.168.120.4):
Custom IP address: http://192.168.120.4:5000
Also enable: “Enable '.m3u8' / '.mpd' link extension”.



That completes the basic setup of Telerising. Under “Watch TV” you can check whether the login works. Then open “M3U List” and copy the link via “Copy” – we will need it in Jellyfin in a moment.

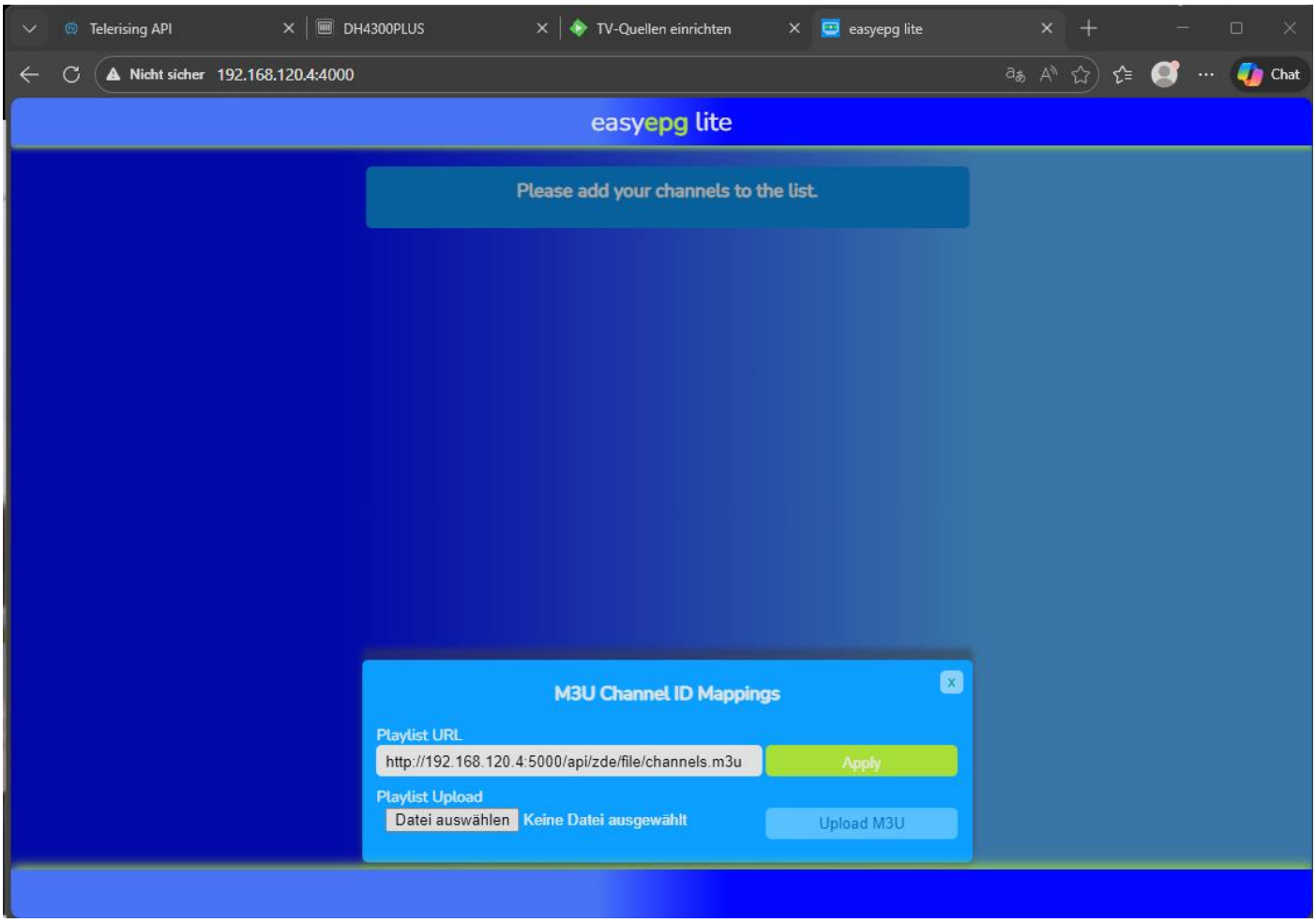


Note: Integration into Jellyfin (M3U tuner + XMLTV EPG + mapping) is done in section 8.

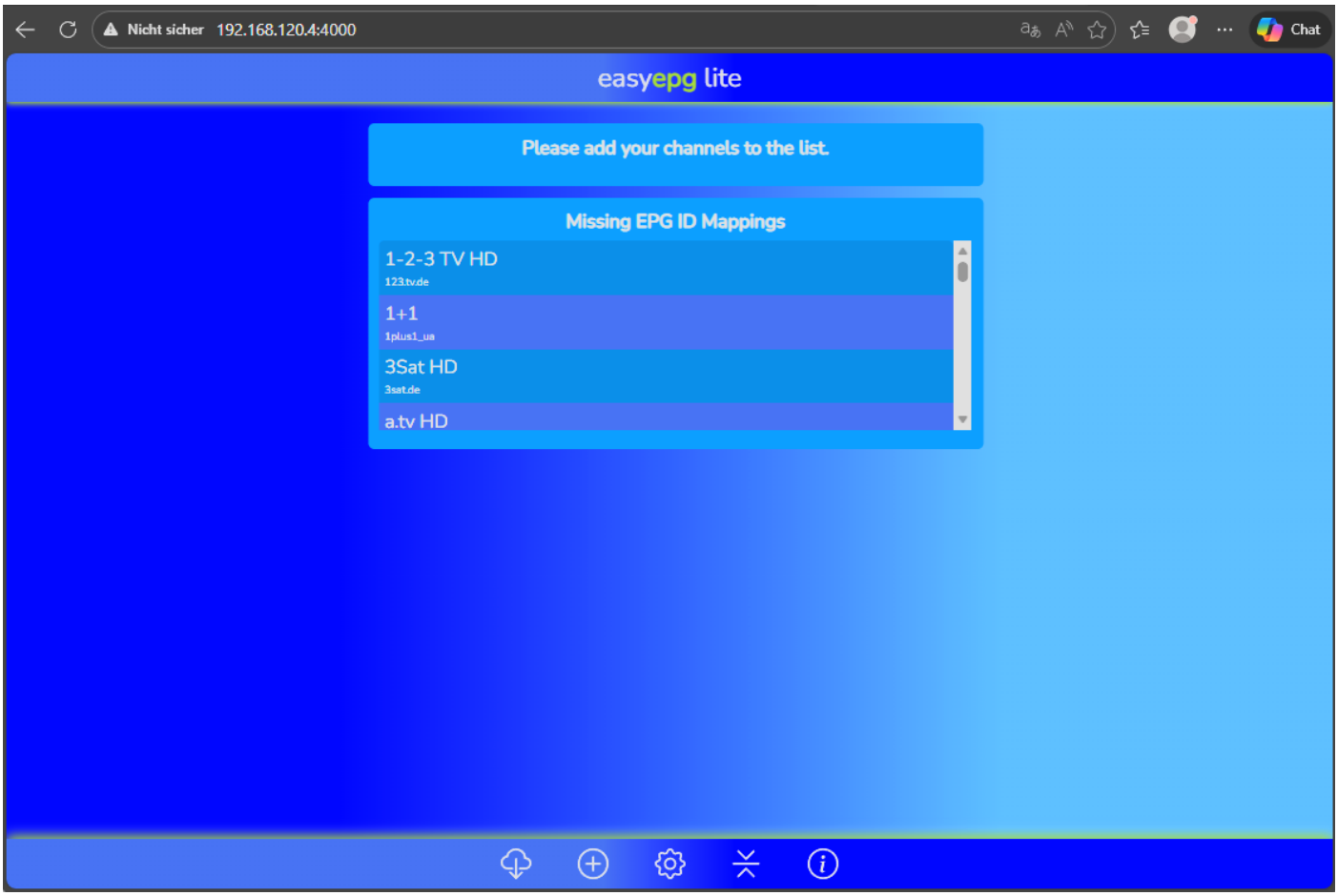
7.4 EasyEPG

Open EasyEPG in your browser at http://NAS-IP:4000/ (replace NAS-IP with the IP address of your NAS).http://NAS-IP:4000/

In the EasyEPG UI, click the fourth icon (M3U Playlist ID Mapping) and paste the copied link from Telerising again:
http://192.168.120.4:5000/api/zde/file/channels.m3u
Then click Apply.http://192.168.120.4:5000/api/zde/file/channels.m3u

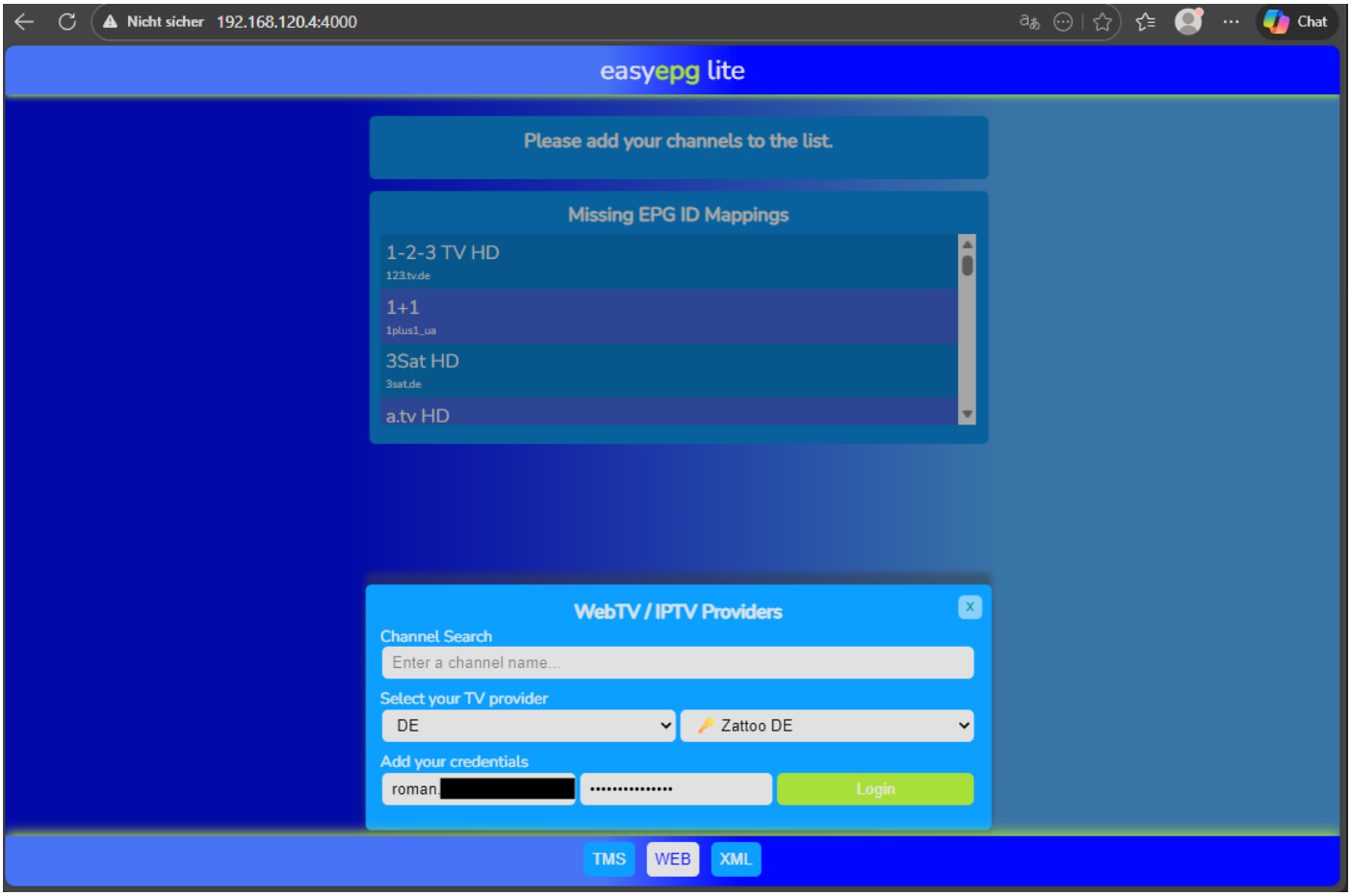


A list of “Missing EPG ID Mappings” will appear.

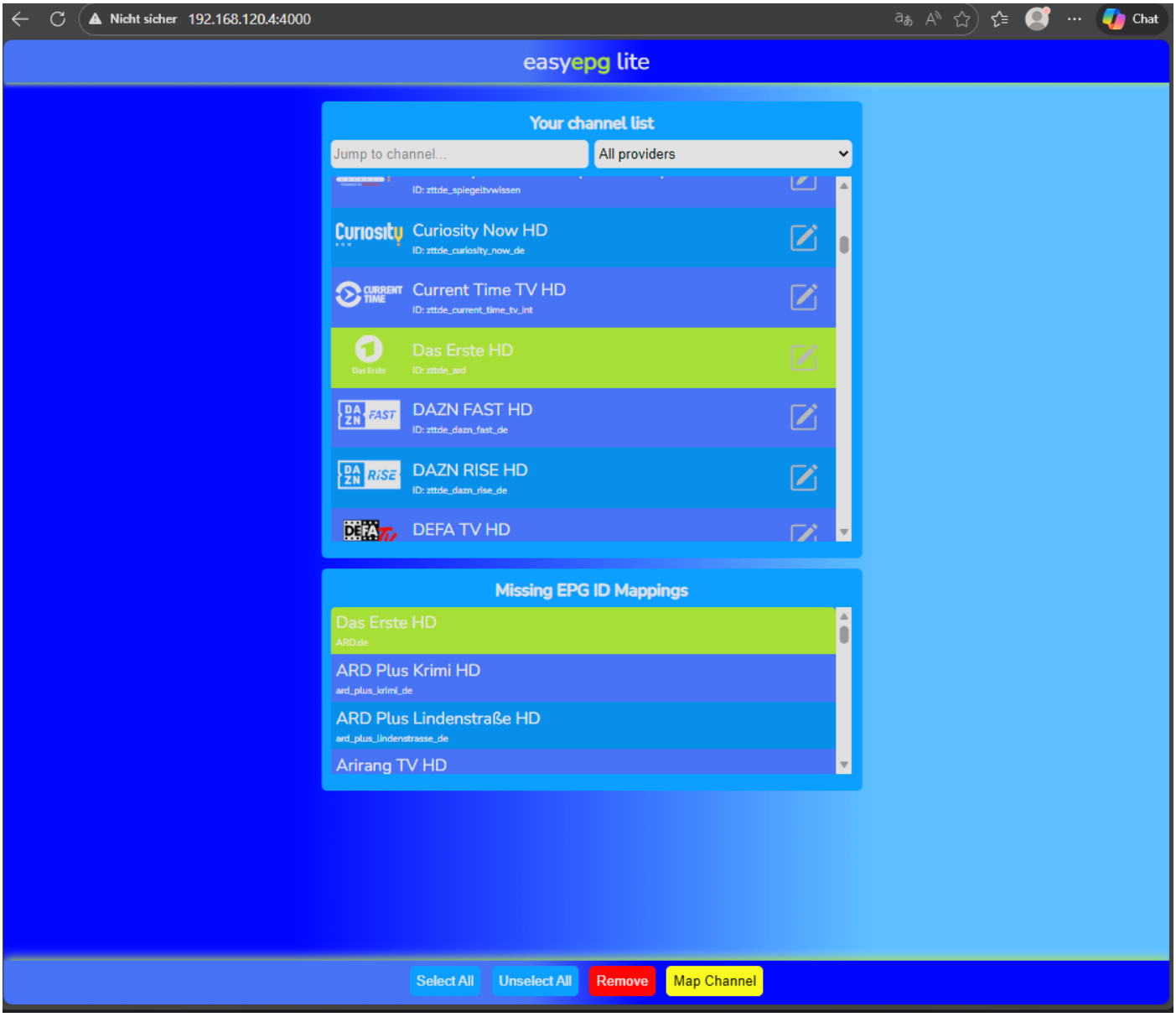


Click the second icon at the bottom left (“Add new Channels”) to assign the missing EPG ID mappings.

In this example I use Zattoo DE: Select the provider, enter your credentials and click “Login”.



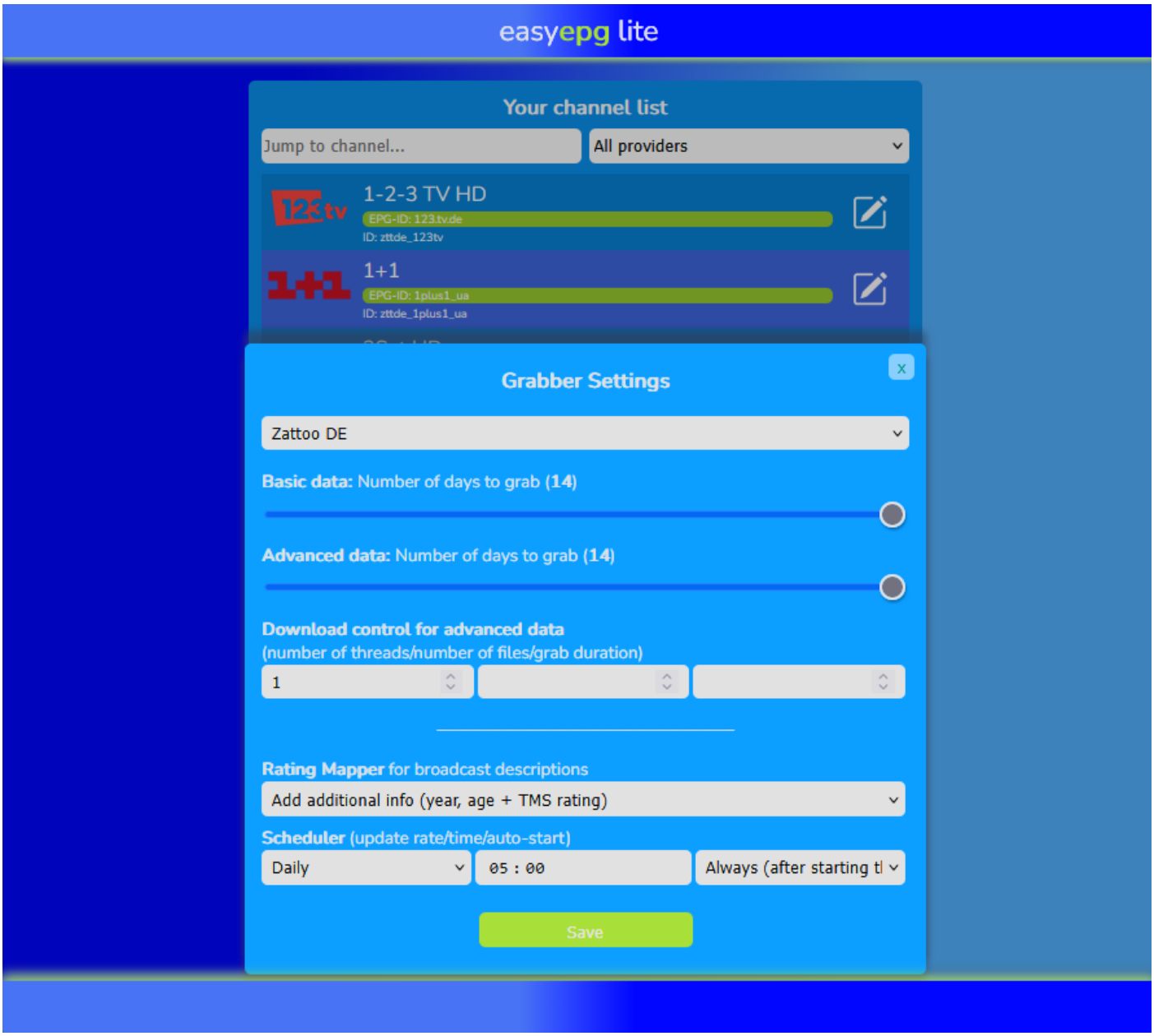
Next, assign suitable EPG data to each channel you want:
Under “Missing EPG ID Mappings” at the bottom, select a channel and then search for it in the list above.
Example: click “Das Erste HD” at the bottom, then select “Das Erste HD” at the top.
Repeat until no “Missing EPG ID Mappings” are shown at the bottom.



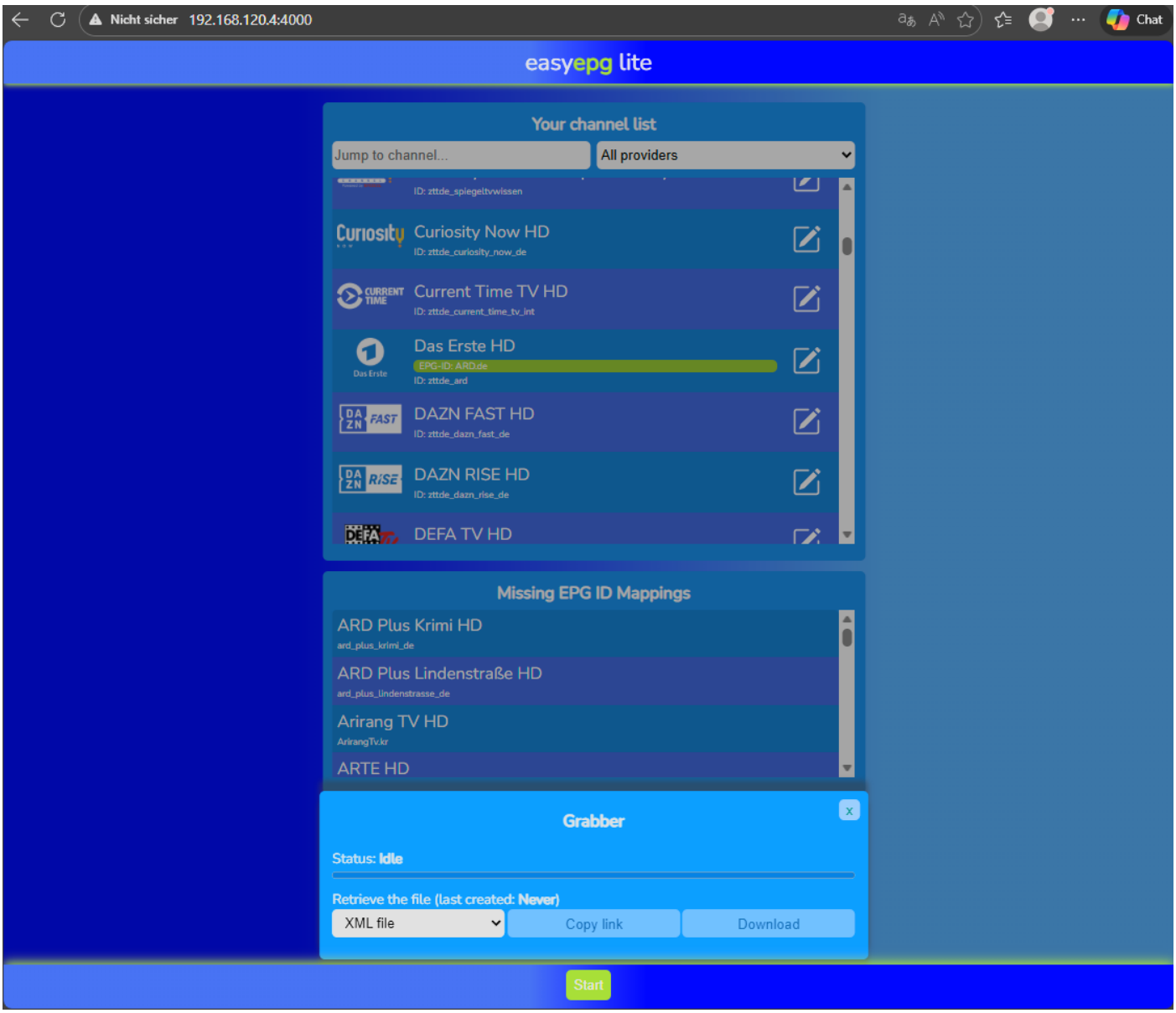
When the mapping is done, click the third icon (“Open Settings”).
Set how many days of EPG data should be loaded.

Recommended settings:
• Rating Mapper: “Add additional info (year, age + TMS rating)”
• Scheduler: “Daily 05:00” and “Always after starting the app” (or your own time)

Then click “Save” and close the grabber settings with “X”.



Finally, click the first icon at the bottom left (“Grabber”), check the settings and start the run with “Start”.
Wait until the status “File created successfully!” appears – only then continue in Jellyfin.

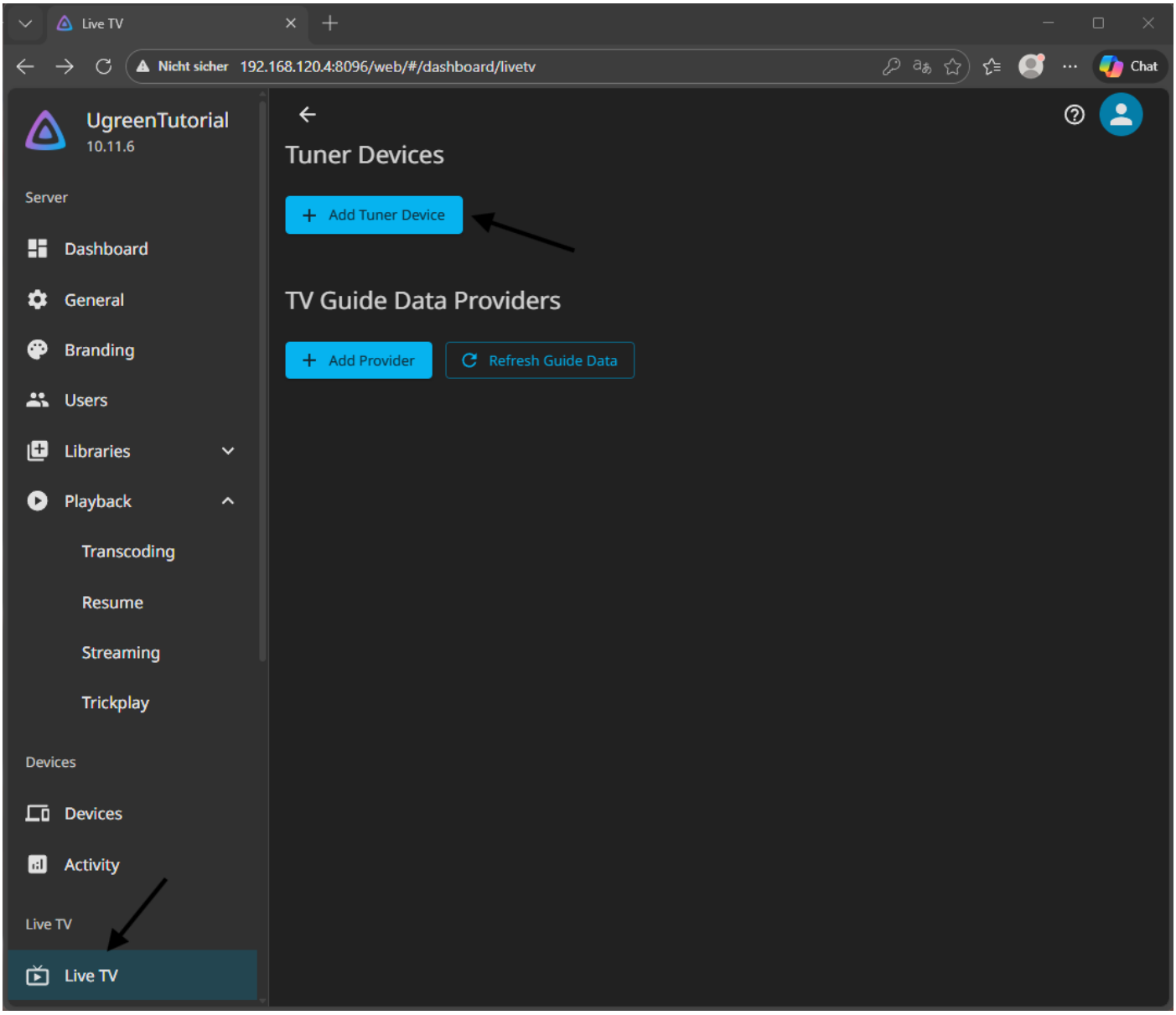


8. Jellyfin Live TV (Telerising + EasyEPG)

The basic configuration of Telerising and EasyEPG is done as in sections 7.3 and 7.4. In this section we connect both in Jellyfin (M3U + XMLTV).

Open Jellyfin and click the admin icon in the top right → Dashboard.

Navigate to “Live TV”.



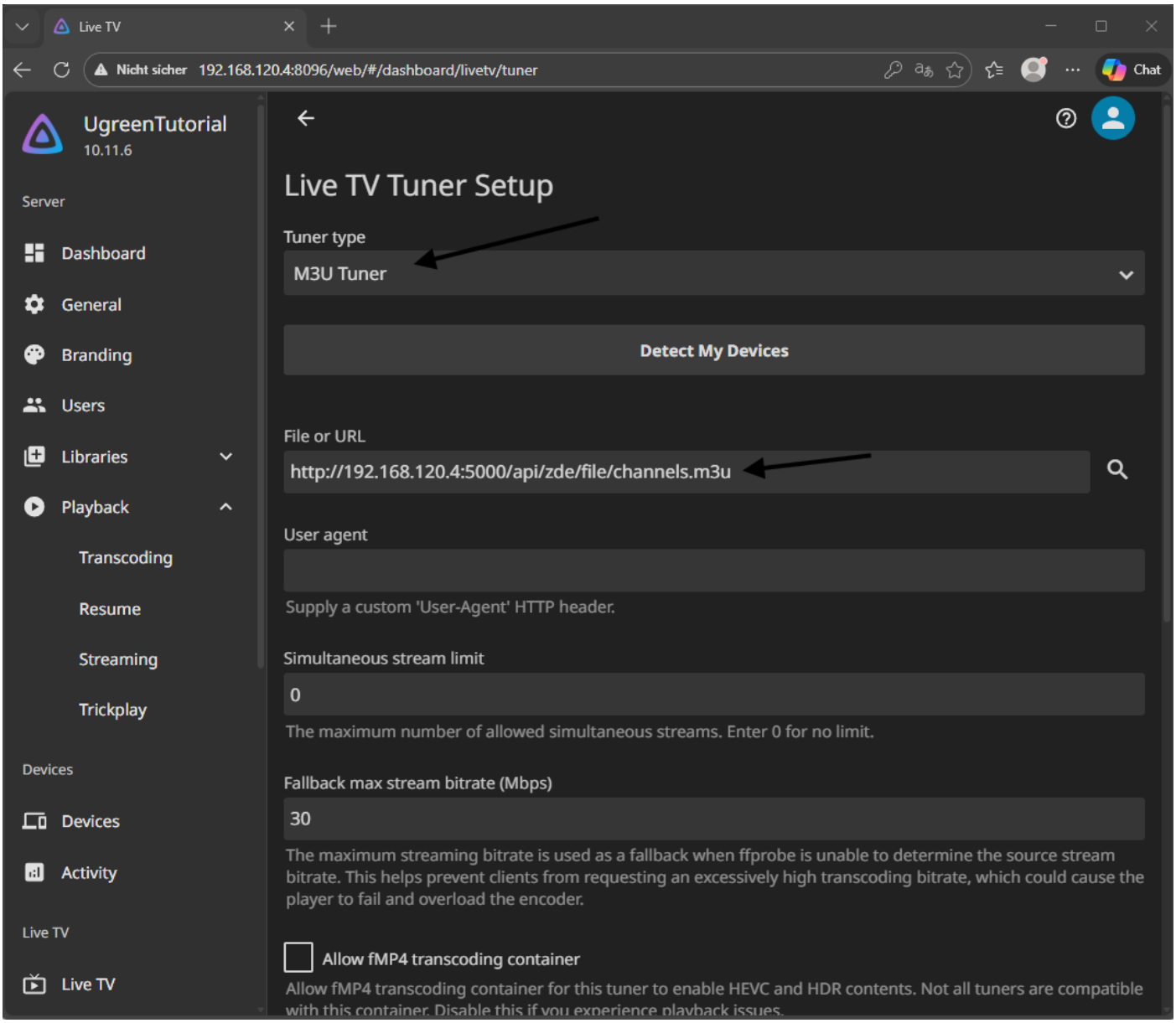
8.1 Add TV tuner (M3U from Telerising)

Under “Add Tuner Devices”, click “+”.

Tuner Type: select “M3U Tuner”.

File or URL: enter the M3U link from Telerising (example: http://192.168.120.4:5000/api/zde/file/channels.m3u).

Save.



8.2 Add EPG (XMLTV from EasyEPG)

Under “TV Guide Data Providers” (or “TV guide data providers”), click “+ Add Provider”.

Provider: select “XMLTV”.

Path to the XMLTV file (inside the Jellyfin container): /config/epg/epg.xml

Note: If the file does not exist yet, start a grabber run in EasyEPG and wait until the status “File created” is shown.

Please adjust the following categories:

Kids categories

Children|Children-music|Children-special|Children-talk|Family|Familie|Holiday-children|Holiday-children special|Kids|Kids & Family|Kids / Animation|Kinder|Kinder & Jugend|Kinder Unterhaltung|Kinderfilm|Kinderfilme|Kindersendung|Kinderserien|Märchen|Zeichentrick

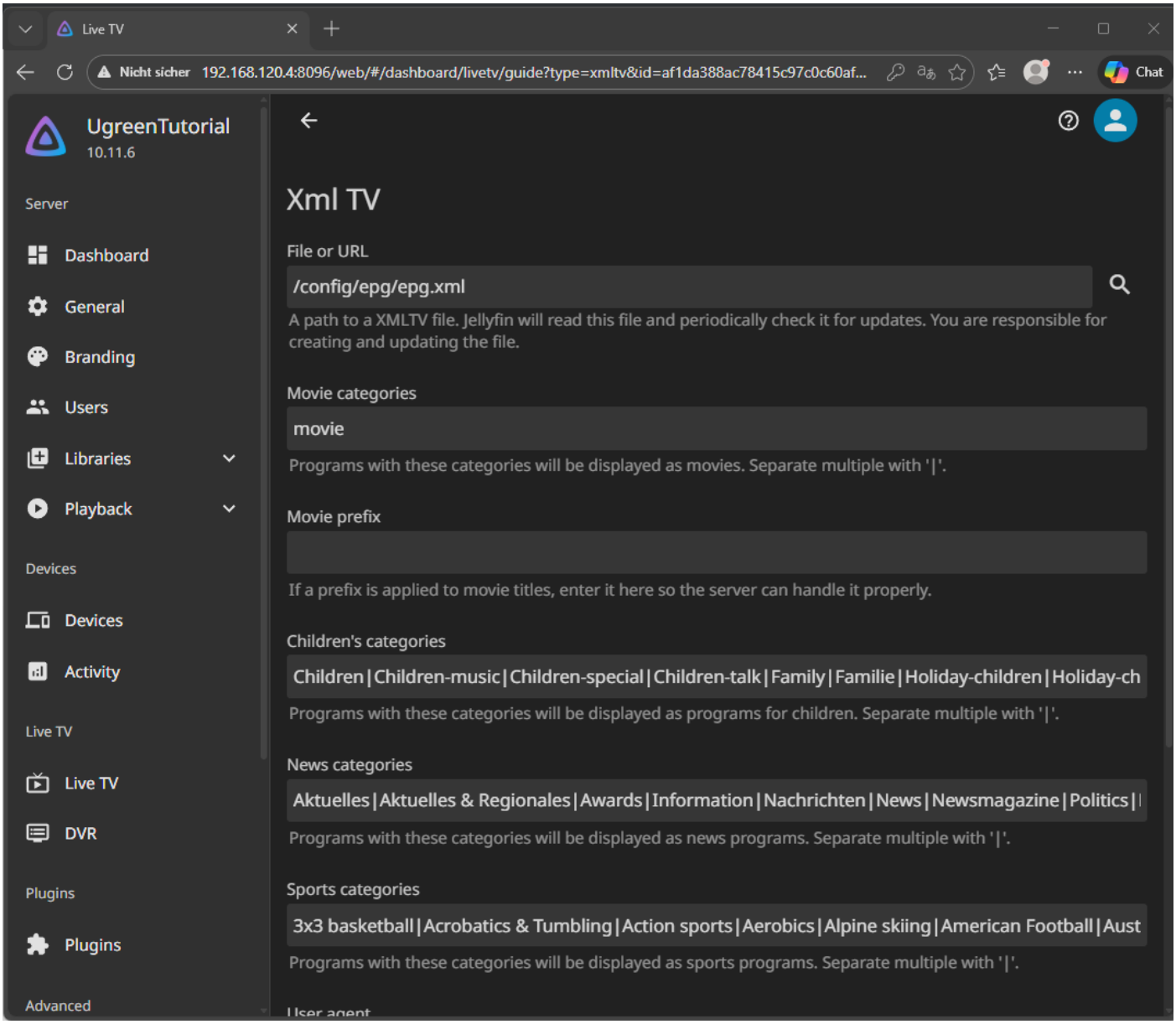
News categories

Aktuelles|Aktuelles & Regionales|Awards|Information|Nachrichten|News|Newsmagazine|Politics|Politik|Regional|Weather|Wetter

Sports category

3x3 basketball|Acrobatics & Tumbling|Action sports|Aerobics|Alpine skiing|American Football|Australian rules football|Auto racing|Badminton|Bandy|Baseball|Basketball|Beach soccer|Beach volleyball|Biathlon|Bicycle|Bicycle racing|Billiards|Billard/Snooker|Boat racing|Boxen|Boxing|Camogie|Canoe|Classic Sport Event|Cricket|Cross-country skiing|Curling|Cycling|Darts|Drag racing|Drift racing|E-Sport|Esports|eSports|Extremsport|Figure skating|Field hockey|Floorball|Football|Fussball|Fußball|Gaelic football|Golf|Gymnastics|Handball|Harness racing|Horse racing|Hydroplane racing|Indoor soccer|Intl basketball|Intl hockey|Intl soccer|Judo|Karate|Kayaking|Laufsport|Leichtathletik|Marathon|Mixed martial arts|Modern pentathlon|Motorcycle racing|Motorsport|Motorsports|Mountain biking|Multi-sport event|Netball|Nordic Combined|Olympics|Paralympics|Playoff sports|Polo|Pool|Pro wrestling|Rhythmic Gymnastics|Road cycling|Roller derby|Rowing|Rugby|Rugby league|Rugby union|Running|Sailing|Sepak takraw|Shinty|Shooting|Short track speed skating|Skateboarding|Skating|Ski Jumping|Skiing|Snowboarding|Snowmobile|Snooker|Softball|Sport|Sportfilm|Sportmagazine|Sportnews|Sport/andere|Sport climbing|Sports|Sports event|Sports non-event|Sports related|Sports talk|Squash|Sumo wrestling|Surfing|Swimming|Swiss Wrestling|Synchronized swimming|Table tennis|Taekwondo|Tanzsport|Tennis|Teqball|Track cycling|Track/field|Trampoline Gymnastics|Triathlon|Turnen|Volleyball|Water polo|Water skiing|Watersports|Weightlifting|Wintersport|Winter Olympics|Wrestling|Yacht racing|Eishockey|Pferdesport|Wassersport

Save.


Live TV
Nicht sicher 192.168.120.4:8096/web/#/dashboard/livetv/guide?type=xmltv&id=af1da388ac78415c97c0c60af...
Chat
UgreenTutorial
10.11.6
Server
Dashboard
General
Branding
Users
Libraries
Playback
Devices
Devices
Activity
Live TV
Live TV
DVR
Plugins
Plugins
Advanced
Xml TV
File or URL
/config/epg/epg.xml
A path to a XMLTV file. Jellyfin will read this file and periodically check it for updates. You are responsible for creating and updating the file.
Movie categories
movie
Programs with these categories will be displayed as movies. Separate multiple with '|'.
Movie prefix
If a prefix is applied to movie titles, enter it here so the server can handle it properly.
Children's categories
Children|Children-music|Children-special|Children-talk|Family|Familie|Holiday-children|Holiday-ch
Programs with these categories will be displayed as programs for children. Separate multiple with '|'.
News categories
Aktuelles|Aktuelles & Regionales|Awards|Information|Nachrichten|News|Newsmagazine|Politics|
Programs with these categories will be displayed as news programs. Separate multiple with '|'.
Sports categories
3x3 basketball|Acrobatics & Tumbling|Action sports|Aerobics|Alpine skiing|American Football|Aust
Programs with these categories will be displayed as sports programs. Separate multiple with '|'.
User agent

8.3 Map channels

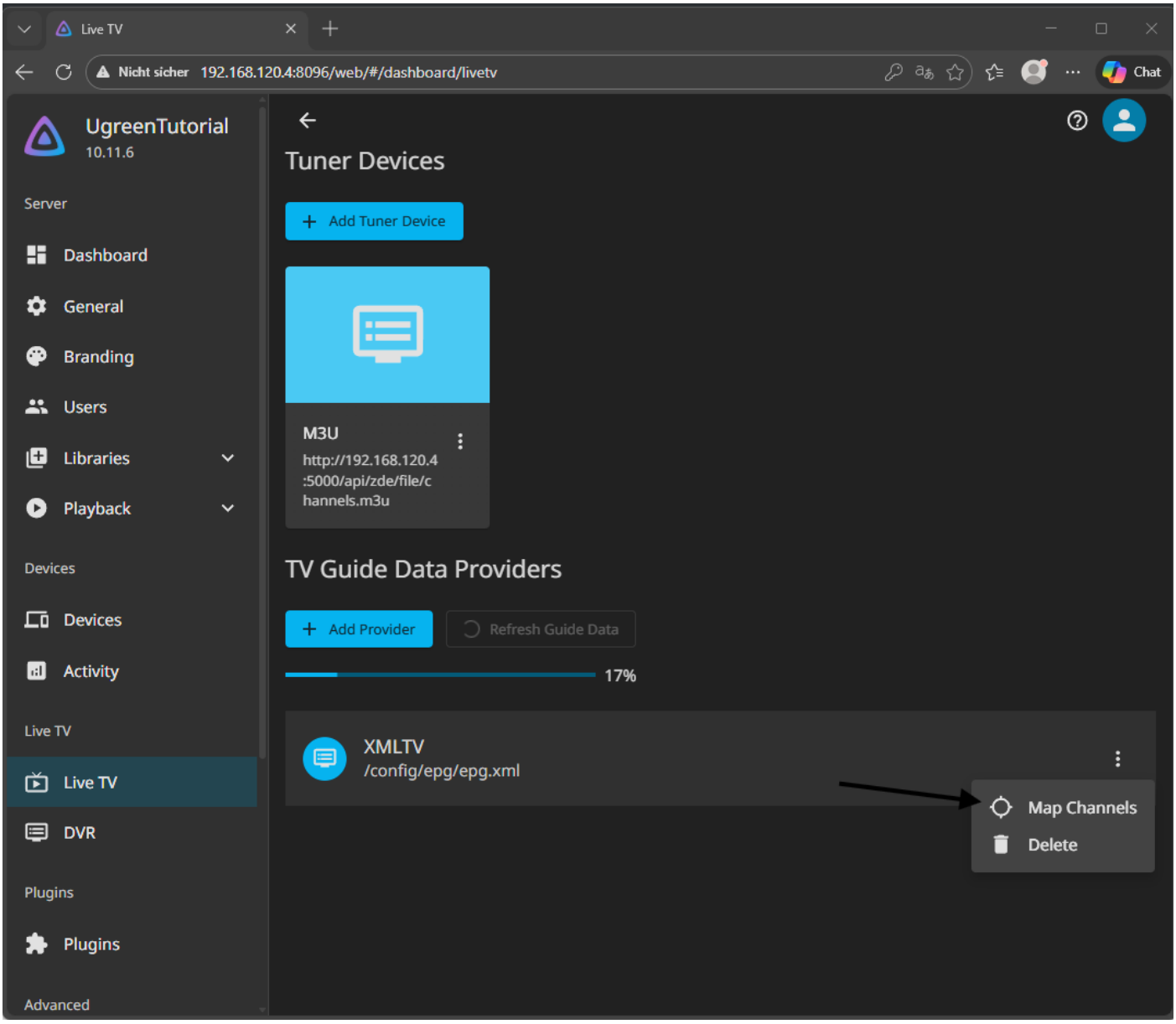
On the guide provider, click "…" → "Map Channels".

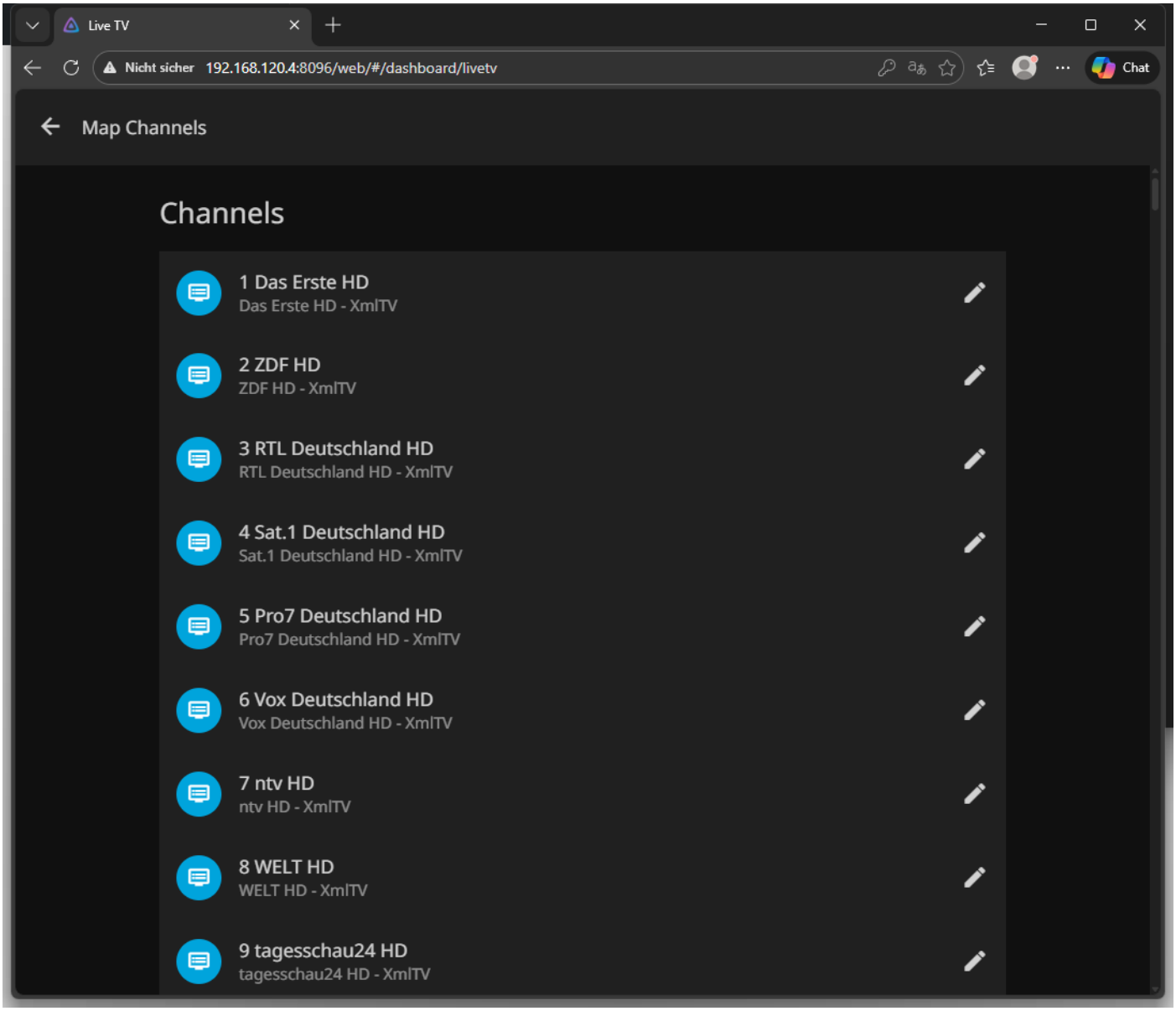
```
/config/epg/epg.xml
```

For each channel, select the matching EPG entry (pencil icon).

Back – after that, the EPG should be imported automatically.

Check: main menu → Live TV → Guide.


Live TV
Nicht sicher 192.168.120.4:8096/web/#/dashboard/livetv
Chat
UgreenTutorial
10.11.6
Server
Dashboard
General
Branding
Users
Libraries
Playback
Devices
Devices
Activity
Live TV
Live TV
DVR
Plugins
Plugins
Advanced
Tuner Devices
Add Tuner Device
M3U
http://192.168.120.4:5000/api/zde/file/channels.m3u
TV Guide Data Providers
Add Provider
Refresh Guide Data
17%
XMLTV
/config/epg/epg.xml
Map Channels
Delete


Live TV
Nicht sicher 192.168.120.4:8096/web/#/dashboard/livetv
Chat
Map Channels
Channels
1 Das Erste HD
Das Erste HD - XmlTV
2 ZDF HD
ZDF HD - XmlTV
3 RTL Deutschland HD
RTL Deutschland HD - XmlTV
4 Sat.1 Deutschland HD
Sat.1 Deutschland HD - XmlTV
5 Pro7 Deutschland HD
Pro7 Deutschland HD - XmlTV
6 Vox Deutschland HD
Vox Deutschland HD - XmlTV
7 ntv HD
ntv HD - XmlTV
8 WELT HD
WELT HD - XmlTV
9 tagesschau24 HD
tagesschau24 HD - XmlTV

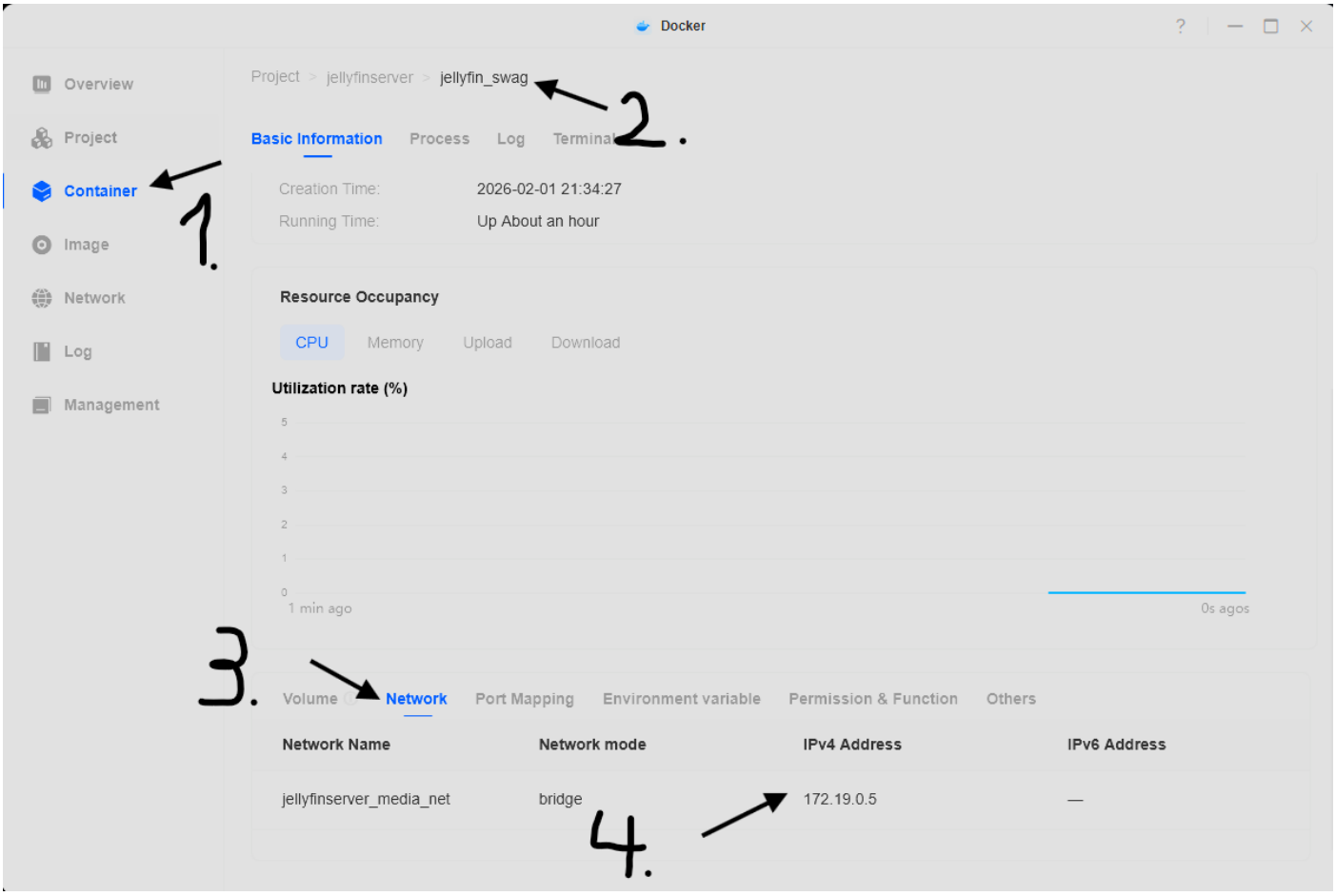
9. Final Jellyfin settings

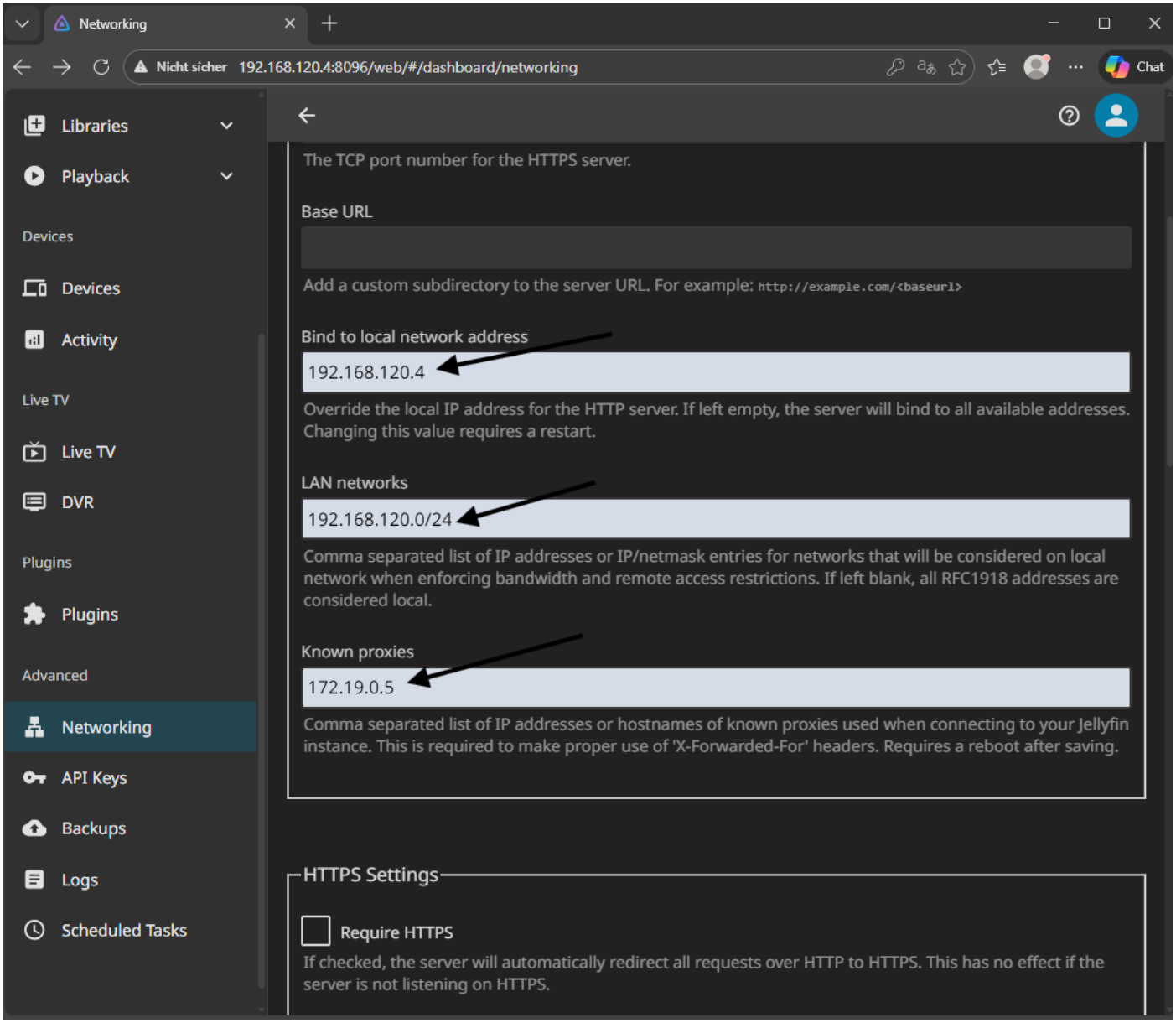
9.1 Reverse proxy / network (SWAG)

If Jellyfin is running behind SWAG, you should add the reverse proxy as a “Known Proxy” so Jellyfin can detect the client IP correctly and rules for external access work properly.

- Dashboard → Network (example):
- Local networks: 192.168.120.0/24 (IP subnet of your router)
- Known Proxies: IP address of the SWAG container in the Docker network
- Optional: If you use additional internal networks (e.g. VPN), add them under “Local networks” as well.

This is how to determine the proxy IP address and then enter it in Jellyfin:
Open the Docker app, select container jellyfin_swag, Network → IPv4 address.
Remember this IP address and enter it under “Known Proxies” in Jellyfin (Network).



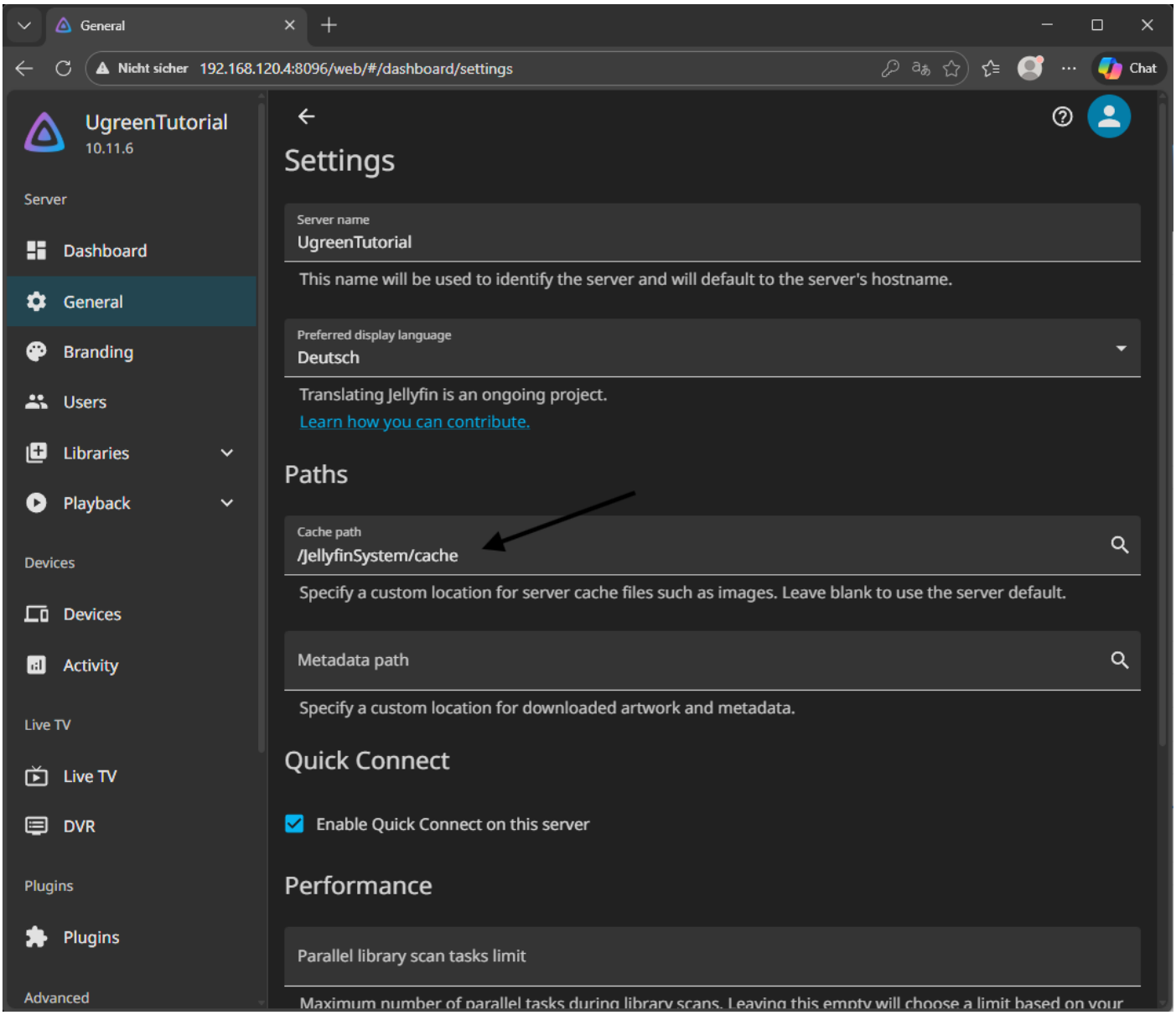

Networking
Nicht sicher 192.168.120.4:8096/web/#/dashboard/networking
Chat
Libraries
Playback
Devices
Devices
Activity
Live TV
Live TV
DVR
Plugins
Plugins
Advanced
Networking
API Keys
Backups
Logs
Scheduled Tasks
The TCP port number for the HTTPS server.
Base URL
Add a custom subdirectory to the server URL. For example: http://example.com/<baseurl>
Bind to local network address
192.168.120.4
Override the local IP address for the HTTP server. If left empty, the server will bind to all available addresses. Changing this value requires a restart.
LAN networks
192.168.120.0/24
Comma separated list of IP addresses or IP/netmask entries for networks that will be considered on local network when enforcing bandwidth and remote access restrictions. If left blank, all RFC1918 addresses are considered local.
Known proxies
172.19.0.5
Comma separated list of IP addresses or hostnames of known proxies used when connecting to your Jellyfin instance. This is required to make proper use of 'X-Forwarded-For' headers. Requires a reboot after saving.
HTTPS Settings
Require HTTPS
If checked, the server will automatically redirect all requests over HTTP to HTTPS. This has no effect if the server is not listening on HTTPS.

9.2 Cache and transcoding paths (JELLYFIN_SYSTEM_DIR)

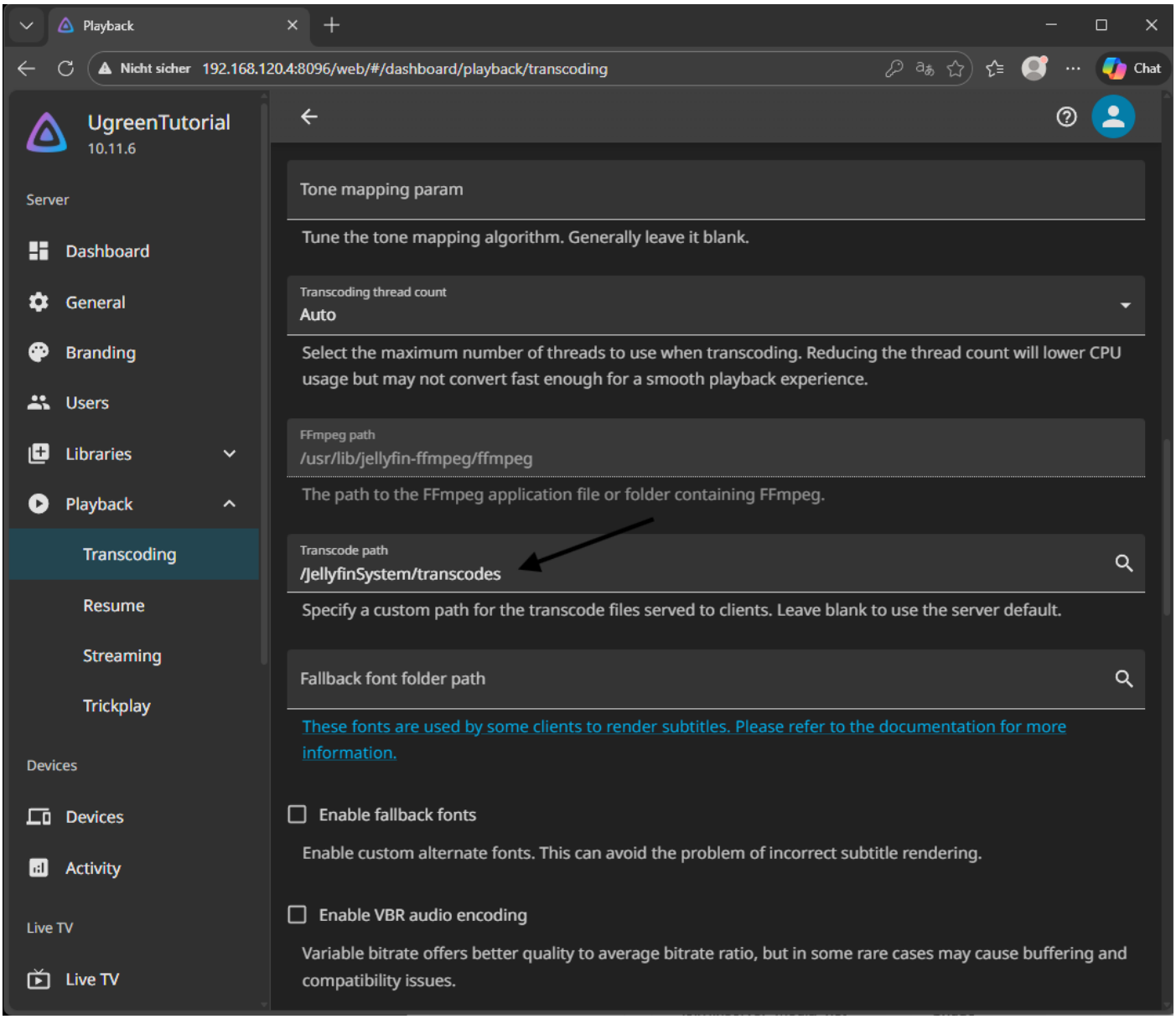
In the compose, ${JELLYFIN_SYSTEM_DIR} is mounted into the container as /JellyfinSystem.

Recommendation (in Jellyfin):

• Administration → General → Paths: Cache path: /JellyfinSystem/cache



• Administration → Playback → Transcoding: Temporary transcoding path: /JellyfinSystem/transcodes



Create folders on the host: ${JELLYFIN_SYSTEM_DIR}/cache and ${JELLYFIN_SYSTEM_DIR}/transcodes.

9.3 Backup (built-in backup/restore)

Jellyfin provides built-in backup/restore. In the dashboard under “Backups” you can create and restore backups.

With the LinuxServer.io Docker image, backups are stored by default under /config/data/data/backups (host: ./Jellyfin/config/data/data/backups).

Tip: In addition, regularly back up the entire host folder ./Jellyfin/config externally (e.g. Sync & Backup / rsync / snapshots).